

Introduction



Magnetism is an interaction that allows certain kinds of objects, which are called ‘magnetic’ objects, to exert forces on each other without physically touching. A magnetic object is surrounded by a magnetic ‘field’ that gets weaker as one moves further away from the object. A second object can feel a magnetic force from the first object because it feels the magnetic field of the first object. The further away the objects are the weaker the magnetic force will be.

See introductory video: (📺 Video: VPfkk at www.everythingscience.co.za)

Humans have known about magnetism for many thousands of years. For example, *lodestone* is a magnetised form of the iron oxide mineral *magnetite*. It has the property of attracting iron objects. It is referred to in old European and Asian historical records; from around 800 BCE in Europe and around 2 600 BCE in Asia.

Magnetic objects stuck to a magnet



Photograph by dougww on Flickr.com

FACT

The root of the English word *magnet* is from the Greek word *magnes*, probably from Magnesia in Asia Minor, once an important source of lodestone.

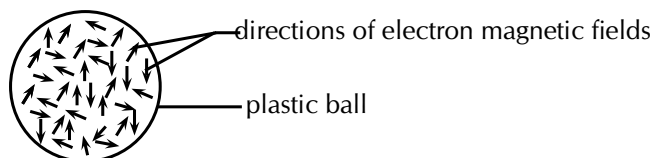
Magnetic fields



A magnetic field is a region in space where a magnet or object made of magnetic material will experience a non-contact, magnetic force.

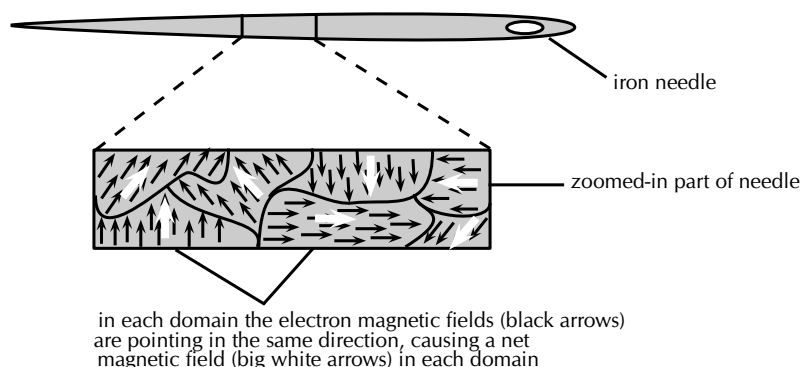
A moving charged particle has a magnetic fields associated with it. One example of a charged particle that occurs in most matter is the electron. Electrons are in constant motion inside material, orbiting the nucleus in the atom which may also be moving, rotating or vibrating.

So electrons inside any object are moving and have magnetic fields associated with them. In most materials these fields point in various directions, so the net magnetic field is zero. For example, in the plastic ball below, the directions of the magnetic fields of the electrons (shown by the arrows) are pointing in different directions and cancel each other out. Therefore the plastic ball is not magnetic and has no magnetic field.



The electron magnetic fields point in all directions
and so there is no net (total) magnetic field for the whole ball

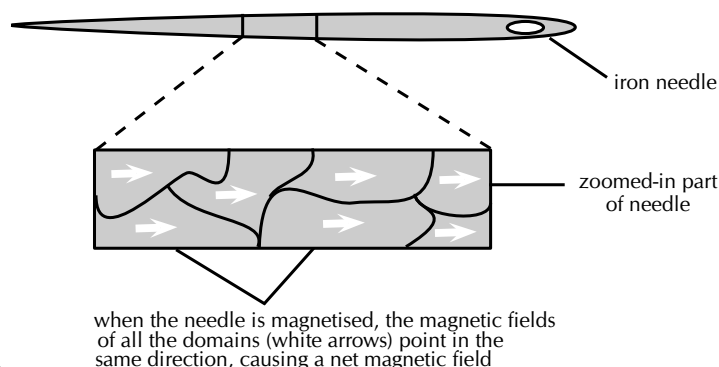
In some materials (e.g. iron), called **ferromagnetic** materials, there are regions called *domains*, where the electrons' magnetic fields line up with each other. All the atoms in each domain are grouped together so that the magnetic fields from their electrons point the same way. The picture shows a piece of an iron needle zoomed in to show the domains with the electric fields lined up inside them.



In permanent magnets, many domains are lined up, resulting in a *net magnetic field*. Objects made from ferromagnetic materials can be magnetised, for example by rubbing a magnet along the object in one direction. This causes the magnetic fields of most, or all, of the domains to line up in one direction. As a result the object as a whole will have a net magnetic field. It is *magnetic*. Once a ferromagnetic object has been magnetised, it can stay magnetic without another magnet being nearby (i.e. without being in another magnetic field). In the picture below, the needle has been magnetised because the magnetic fields in all the domains are pointing in the same direction.

A permanent magnet

Photograph by Harsh Patel



Investigation: *Ferromagnetic materials and magnetisation*

1. Find 2 paper clips. Put the paper clips close together and observe what happens.
 - a. What happens to the paper clips?
 - b. Are the paper clips magnetic?
2. Now take a permanent bar magnet and rub it once along 1 of the paper clips. Remove the magnet and put the paper clip which was touched by the magnet close to the other paper clip and observe what happens. Does the untouched paper clip experience a force on it? If so, is the force attractive or repulsive?
3. Rub the same paper clip a few more times with the bar magnet, in the same direction as before. Put the paper clip close to the other one and observe what happens.
 - a. Is there any difference to what happened in step 2?
 - b. If there is a difference, what is the reason for it?
 - c. Is the paper clip which was rubbed repeatedly by the magnet now magnetised?
 - d. What is the difference between the two paper clips at the level of their atoms and electrons?
4. Now, find a *metal* knitting needle, or a metal ruler, or other metal object. Rub the bar magnet along the knitting needle a few times in the same direction. Now put the knitting needle close to the paper clips and observe what happens.
 - a. Does the knitting needle attract the paper clips?

- b. What does this tell you about the material of the knitting needle? Is it ferromagnetic?
5. Repeat this experiment with objects made from other materials. Which materials appear to be ferromagnetic and which are not? Put your answers in a table.

A ferromagnetic material is a substance that shows spontaneous magnetisation. Spontaneous means self-generated or to happen without external cause. This means that a ferromagnetic material has a magnetic field without any external factors being required.

Permanent magnets

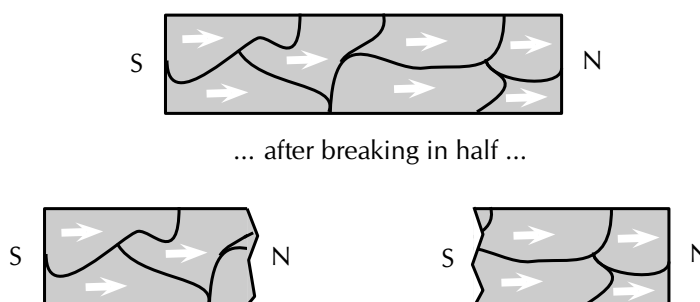


The poles of permanent magnets



Because the domains in a permanent magnet all line up in a particular direction, the magnet has a pair of opposite poles, called **north** (usually shortened to **N**) and **south** (usually shortened to **S**). Even if the magnet is cut into tiny pieces, each piece will still have *both* a N and a S pole. These magnetic poles *a/ways* occur in pairs. In nature, we never find a north magnetic pole or south magnetic pole on its own.

▶ See video: VPflm at www.everythingscience.co.za



In nature, positive and negative electric charges can be found on their own, but you *never* find just a north magnetic pole or south magnetic pole on its own. On the very small scale, zooming in to the size of atoms, magnetic fields are caused by moving charges (i.e. the negatively charged electrons).

Magnetic attraction and repulsion



Like (identical) poles of magnets repel one another whilst unlike (opposite) poles attract. This means that two N poles or two S poles will push away from each other while a N pole and a S pole will be drawn towards each other.

Do you think the following magnets will repel or be attracted to each other?



We are given two magnets with the N pole of one approaching the N pole of the other. Since both poles are the same, the magnets will repel each other.



We are given two magnets with the N pole of one approaching the S pole of the other. Since both poles are different, the magnets will be attracted to each other.

Representing magnetic fields

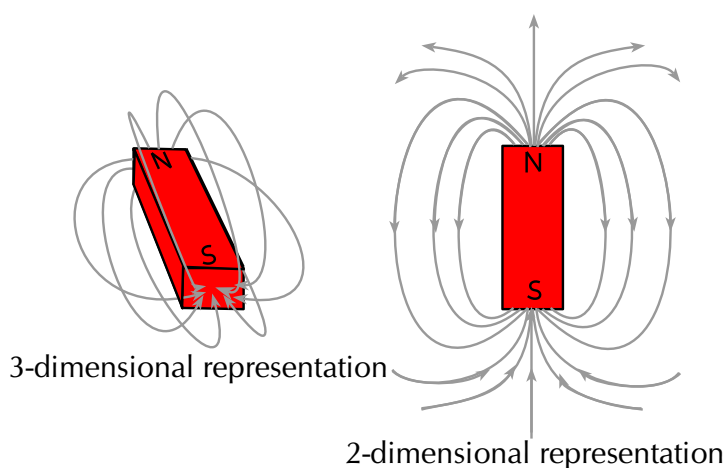


Magnetic fields can be *represented* using **magnetic field lines** starting at the North pole and ending at the South pole. Although the magnetic field of a permanent magnet is everywhere surrounding the magnet (in all three dimensions), we draw only some of the field lines to represent the field (usually only a two-dimensional cross-section is shown in drawings).

▶ See video: VPfmc at www.everythingscience.co.za

Tip

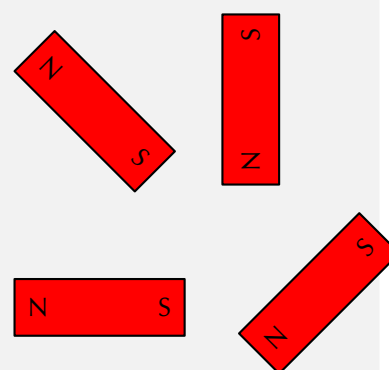
1. Field lines *never* cross.
2. Arrows drawn on the field lines indicate the direction of the field.
3. A magnetic field points from the north to the south pole of a magnet.



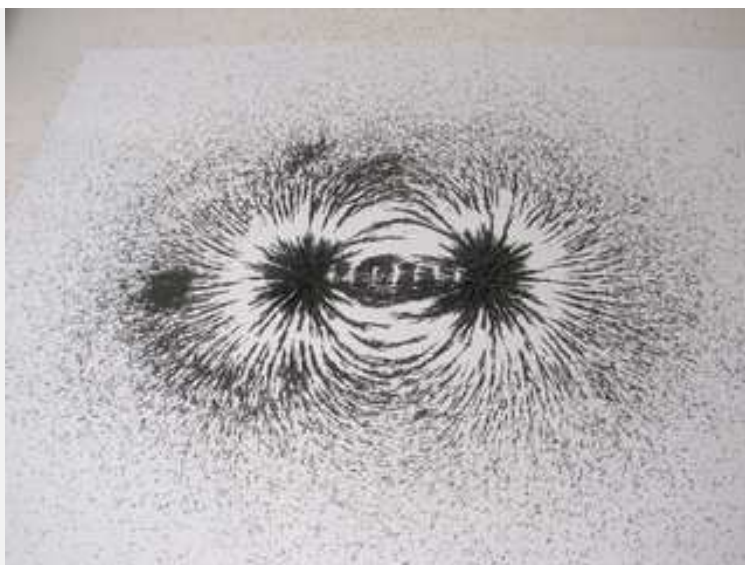
In areas where the magnetic field is strong, the field lines are closer together. Where the field is weaker, the field lines are drawn further apart. The number of field lines drawn crossing a given two-dimensional surface is referred to as the **magnetic flux**. The magnetic flux is used as a measure of the strength of the magnetic field through that surface.

Investigation: Magnetic field around a bar magnet

Take a bar magnet and place it under a non-magnetic, thin flat surface (this is to stop the paper bending). Place a sheet of white paper on the surface over the bar magnet and sprinkle some iron filings onto the paper. Give the paper a shake to evenly distribute the iron filings. In your workbook, draw the bar magnet and the pattern formed by the iron filings. Draw the pattern formed when you rotate the bar magnet to a different angle as shown alongside.



Iron filings revealing a magnetic field

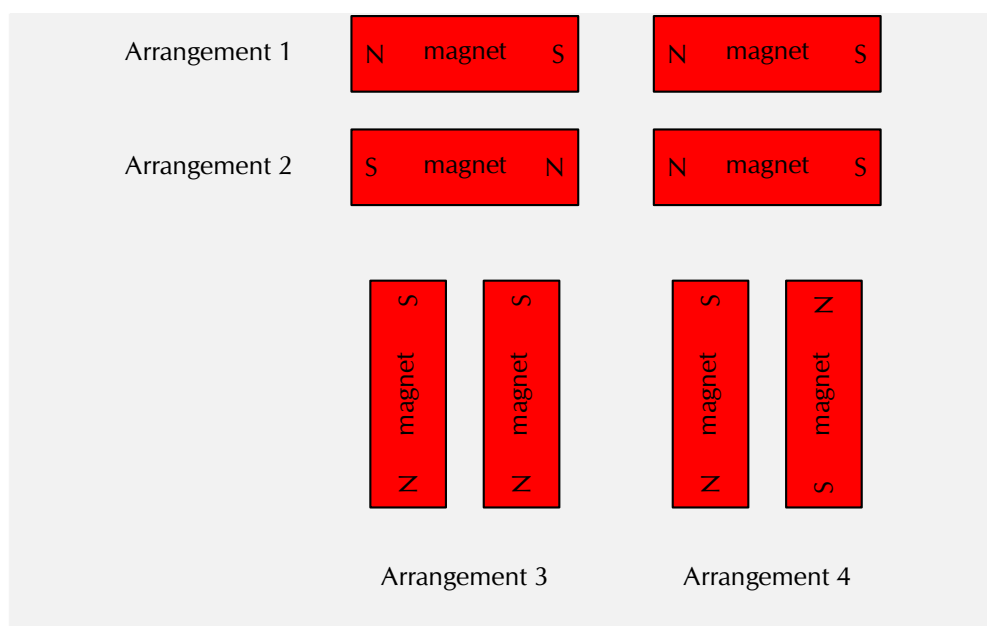


Photograph by oskay on Flickr.com

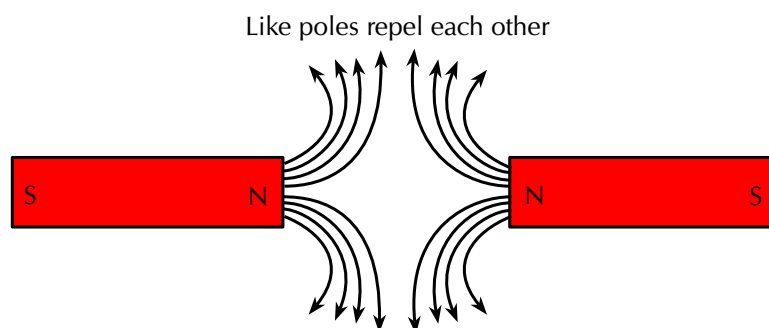
As the investigation shows, one can map the magnetic field of a magnet by placing it underneath a piece of paper and sprinkling iron filings on top. The iron filings line themselves up parallel to the magnetic field.

Investigation: *Magnetic field around a pair of bar magnets*

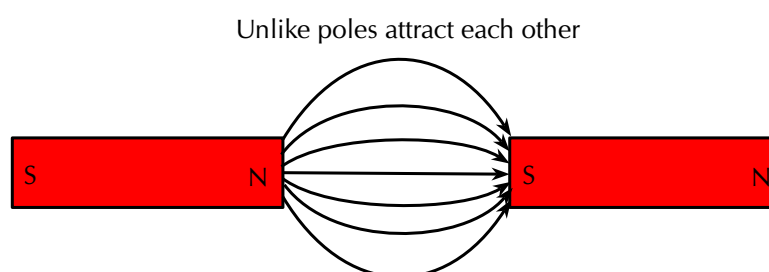
Take two bar magnets and place them a short distance apart such that they are repelling each other. Place a sheet of white paper over the bar magnets and sprinkle some iron filings onto the paper. Give the paper a shake to evenly distribute the iron filings. In your workbook, draw both the bar magnets and the pattern formed by the iron filings. Repeat the procedure for two bar magnets attracting each other and draw what the pattern looks like for this situation. Make a note of the shape of the lines formed by the iron filings, as well as their size and their direction for both arrangements of the bar magnet. What does the pattern look like when you place both bar magnets side by side?



As already stated, opposite poles of a magnet attract each other and bringing them together causes their magnetic field lines to *converge* (come together). Like poles of a magnet repel each other and bringing them together causes their magnetic field lines to *diverge* (bend out from each other).



The field lines between 2 like poles diverge



The magnetic field lines between 2 unlike poles converge

Ferromagnetism



Ferromagnetism is a phenomenon shown by materials like iron, nickel or cobalt. These materials can form permanent magnets. They always magnetise so as to be attracted to a magnet, no matter which magnetic pole is brought toward the unmagnetised iron/nickel/cobalt.

Retentivity and magnetic materials [NOT IN CAPS]



The ability of a ferromagnetic material to retain its magnetisation *after* an external field is removed is called its **retentivity**.

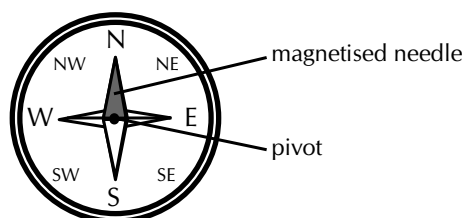
Paramagnetic materials are materials like aluminium or platinum, which become magnetised in an external magnetic field in a similar way to ferromagnetic materials. However, they lose their magnetism when the external magnetic field is removed.

Diamagnetism is shown by materials like copper or bismuth, which become magnetised in a magnetic field with a polarity *opposite* to the external magnetic field. Unlike iron, they are slightly repelled by a magnet.

The compass

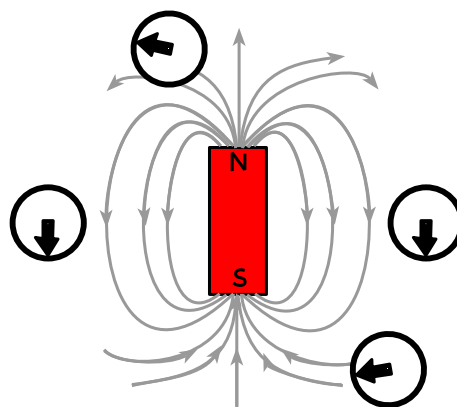


A **compass** is an instrument which is used to find the direction of a magnetic field. A compass consists of a small metal needle which is magnetised itself and which is free to turn in any direction. Therefore, when in the presence of a magnetic field, the needle is able to line up in the same direction as the field.



FACT

Lodestone, a magnetised form of iron-oxide, was found to orientate itself in a north-south direction if left free to rotate by suspension on a string or on a float in water. Lodestone was therefore used as an early navigational compass.



The direction of the compass arrow is the same as the direction of the magnetic field

Compasses are mainly used in navigation to find direction on the earth. This works because the Earth itself has a magnetic field which is similar to that of a bar magnet (see the picture below). The compass needle aligns with the Earth's magnetic field direction and points north-south. Once you know where north is, you can figure out any other direction. A photograph of a compass is shown on the right.

Some animals can detect magnetic fields, which helps them orientate themselves and navigate. Animals which can do this include pigeons, bees, Monarch butterflies, sea turtles and certain fish.

A compass



Photograph by Colin Zhu

The Earth's magnetic field

 **ESAEO**

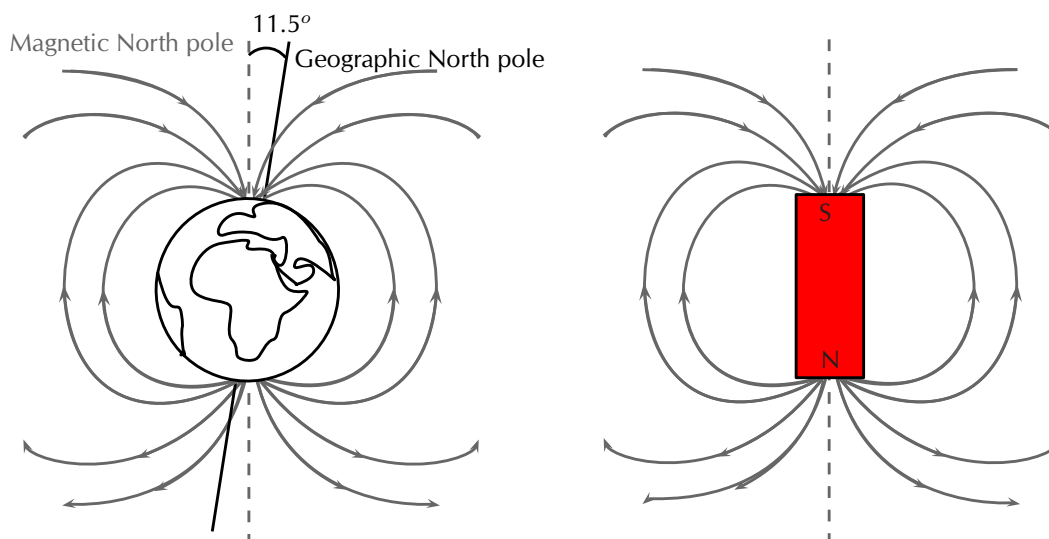
In the picture below, you can see a representation of the Earth's magnetic field which is very similar to the magnetic field of a giant bar magnet like the one on the right of the picture. The Earth has two **magnetic poles**, a north and a south pole just like a bar magnet.

In addition to the magnetic poles the Earth also has two **geographic poles**. The two geographic poles are the points on the Earth's surface where the line of the Earth's axis of rotation meets the surface. To visualise this you could take any round fruit (lemon, orange etc.) and stick a pencil through the middle so it comes out the other side. Turn the pencil, the pencil is the axis of rotation and the geographic poles are where the pencil enters and

exits the fruit. We call the geographic north pole **true north**.

The Earth's magnetic field has been measured very precisely and scientists have found that the magnetic poles do not correspond exactly to the geographic poles.

So the Earth has two north poles and two south poles: **geographic poles** and **magnetic poles**.



FACT

The direction of the Earth's magnetic field flips direction about once every 200 000 years! You can picture this as a bar magnet whose north and south pole periodically switch sides. The reason for this is still not fully understood.

The Earth's magnetic field is thought to be caused by flowing liquid metals in the outer core of the planet which causes electric currents and a magnetic field. From the picture you can see that the direction of magnetic north and true north are not identical. The **geographic north pole** is about $11,5^\circ$ away from the direction of the **magnetic north pole** (which is where a compass will point). However, the magnetic poles shift slightly all the time.

Another interesting thing to note is that if we think of the Earth as a big bar magnet, and we know that magnetic field lines always point *from north to south*, then the compass tells us that what we call the *magnetic north pole* is actually the *south pole* of the bar magnet!

Phenomena related to the Earth's magnetic field

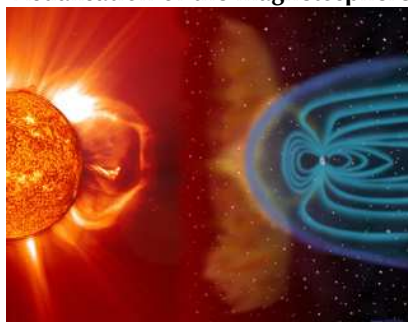


The importance of the magnetic field to life on Earth

The Earth's magnetic field is very important for humans and other animals on Earth because it protects us from being bombarded (hit) by high energy charged particles which are emitted by the Sun. The stream of charged particles (mainly positively charged protons and negatively charged electrons) coming from the sun is called the solar wind. When

these particles come close to the Earth, they are deflected by the Earth's magnetic field and cannot shower down to the surface where they can harm living organisms. Astronauts in space are at risk of being irradiated by the solar wind because they are outside the zones where the charged particles are trapped.

Visualisation of the magnetosphere



*Photo by NASA Goddard Photo and Video
on Flickr.com*

The region above Earth's atmosphere in which charged particles are affected the Earth's magnetic field is called the magnetosphere. Relatively often, in addition to the usual solar wind, the Sun may eject a large bubble of material (protons and electrons) with its own magnetic field from its outer atmosphere. Sometimes these bubbles travel towards the Earth where their magnetic fields can join with Earth's magnetic field. When this happens a huge amount of energy is released into the Earth's magnetosphere, causing a geomagnetic storm. These storms cause rapid changes in the Earth's magnetosphere which in turn may affect electric and magnetic systems on the Earth such as power grids, cellphone networks, and other electronic systems.

Aurorae (pronounced Or-roar-ee)

Another effect caused by the Earth's magnetic field is the spectacular Northern and Southern Lights, which are also called the Aurora Borealis and the Aurora Australis respectively.

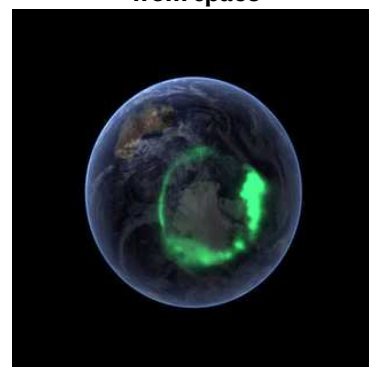
When charged particles from the solar wind reach the Earth's magnetosphere, they spiral along the magnetic field lines towards the North and South poles. If they collide with particles in the Earth's atmosphere, they can cause red or green lights which stretch across a large part of the sky and which is called the aurora.

Aurora borealis photographed in Alaska



Photograph by Trodel on Flickr.com

Aurora australis photographed from space



*Photograph by seishin17 on
Flickr.com*

As this only happens close to the North and South poles, we cannot see the aurorae from South Africa. However, people living in the high Northern latitudes in Canada, Sweden, and Finland, for example, often see the Northern lights.

See simulation: (📺 Simulation: VPfkg at www.everythingscience.co.za)

Chapter 15 | Summary

See the summary presentation (📺 Presentation: VPfjh at www.everythingscience.co.za)

- Magnets have two poles - North and South.
- Some substances can be easily magnetised.
- Like poles repel each other and unlike poles attract each other.
- The Earth also has a magnetic field.
- A compass can be used to find the magnetic north pole and help us find our direction.
- The Earth's magnetic field protects us from being bombarded by high energy charged particles which are emitted by the Sun.
- The Aurorae are an effect of the Earth's magnetic field.

Chapter 15

End of chapter exercises

1. Describe what is meant by the term *magnetic field*.
2. Use words and pictures to explain why permanent magnets have a magnetic field around them. Refer to *domains* in your explanation.
3. What is a magnet?
4. What happens to the poles of a magnet if it is cut into pieces?
5. What happens when like magnetic poles are brought close together?
6. What happens when unlike magnetic poles are brought close together?

7. Draw the shape of the magnetic field around a bar magnet.
8. Explain how a compass indicates the direction of a magnetic field.
9. Compare the magnetic field of the Earth to the magnetic field of a bar magnet using words and diagrams.
10. Explain the difference between the geographical north pole and the magnetic north pole of the Earth.
11. Give examples of phenomena that are affected by Earth's magnetic field.
12. Draw a diagram showing the magnetic field around the Earth.

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 005z (2.) 0060 (3.) 0061 (4.) 0062 (5.) 0063 (6.) 0064
(7.) 0065 (8.) 0066 (9.) 0067 (10.) 0068 (11.) 0069 (12.) 006a

Introduction and key concepts



Electrostatics is the study of electric charge which is at rest or static (not moving). In this chapter we will look at some of the basic principles of electrostatics as well as the principle of conservation of charge.

See introductory video: (▶ Video: VPfmg at www.everythingscience.co.za)

Two kinds of charge

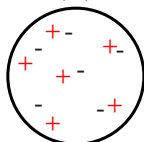


All objects surrounding us (including people!) contain large amounts of electric charge. There are two types of electric charge: **positive** charge and **negative** charge. If the same amounts of negative and positive charge are found in an object, there is *no net charge* and the object is electrically **neutral**. If there is more of one type of charge than the other on the object then the object is said to be **electrically charged**. The picture below shows what the distribution of charges might look like for a neutral, positively charged and negatively charged object.

There are:

6 positive charges and
6 negative charges

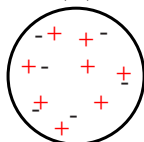
$$6 + (-6) = 0$$



There is zero net charge:
The object is neutral

8 positive charges and
6 negative charges

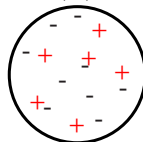
$$8 + (-6) = 2$$



The net charge is +2
The object is positively charged

6 positive charges and
9 negative charges

$$6 + (-9) = -3$$



The net charge is -3

Positive charge is carried by the protons in material and negative charge by electrons. The overall charge of an object is usually due to changes in the number of electrons. To make an object:

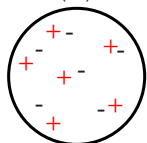
- **Positively charged:** electrons are removed making the object electron **deficient**.
- **Negatively charged:** electrons are added giving the object an **excess** of electrons.

So in practise what happens is that the number of positive charges (protons) remains the same and the number of electrons changes:

There are:

6 positive charges and
6 negative charges

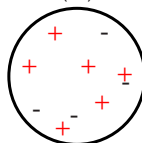
$$6 + (-6) = 0$$



There is zero net charge:
The object is neutral

6 positive charges and
4 negative charges

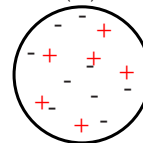
$$6 + (-4) = 2$$



The net charge is +2
The object is positively charged

6 positive charges and
9 negative charges

$$6 + (-9) = -3$$



The net charge is -3
The object is negatively charged

Tribo-electric charging



Tip

Charge, like energy, cannot be created or destroyed. We say that charge is **conserved**.

Objects may become charged in many ways, including by contact with or being rubbed by other objects. This means that they can gain or lose negative charge. For example, charging happens when you rub your feet against the carpet. When you then touch something metallic or another person, you feel a shock as the excess charge that you have collected is *discharged*.

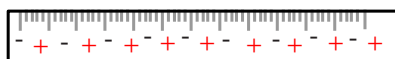
When you rub your feet against the carpet, negative charge is transferred to you from the carpet. The carpet will then become positively charged by the *same amount*.

▶ See video: VPfns at www.everythingscience.co.za

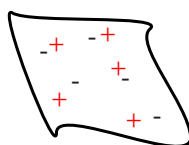
Another example is to take two *neutral* objects such as a plastic ruler and a cotton cloth (handkerchief). To begin, the two objects are neutral (i.e. have the same amounts of positive and negative charge).

Note: We represent the positive charge with a + and the negative charge with a -. This is just to **illustrate the balance and changes that occur, not the actual location** of the positive and negative charges. The charges are spread throughout the material and the real change happens by increasing or decreasing electrons on the surface of the materials.

BEFORE rubbing:



The ruler has 9 positive charges and 9 negative charges

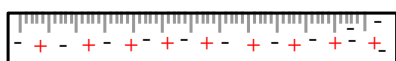


The neutral cotton cloth has 5 positive charges and 5 negative charges

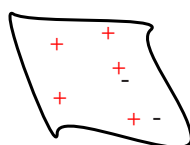
The total number of charges is:
 $(9 + 5) = 14$ positive charges
 $(9 + 5) = 14$ negative charges

Now, if the cotton cloth is used to rub the ruler, negative charge is transferred *from* the cloth *to* the ruler. The ruler is now *negatively* charged (i.e. has an excess of electrons) and the cloth is *positively* charged (i.e. is electron deficient). If you count up all the positive and negative charges at the beginning and the end, there are still the same amount, i.e. total charge has been *conserved*!

AFTER rubbing:



The ruler has 9 positive charges and 12 negative charges
 It is now negatively charged.



The cotton cloth has 5 positive charges and 2 negative charges.
 It is now positively charged.

The total number of charges is:
 $(9+5)=14$ positive charges
 $(12+2)=14$ negative charges

Charges have been transferred from the cloth to the ruler BUT total charge has been conserved!

Example 1: Tribo-electric charging

QUESTION

If you have a cotton cloth and a silk cloth and you rub them together, which becomes negatively charged?

SOLUTION

Step 1 : Analyse the information provided

There are two materials provided and they will be rubbed together. This means we are dealing with the interaction between the materials. The question is related to the charge on the materials which we can assume were neutral to begin with. This means that we are dealing with electrostatics and the interaction of materials leading to the materials

becoming charged is tribo-electric charging.

Step 2 : *Extract material properties*

Locate the materials in the tribo-electric series. The key thing is to know which is more positive and more negative in the series. Silk falls above cotton in our table making it more positive in the series.

Step 3 : *Apply principles*

We know that when two materials are rubbed the more negative one in the series gains electrons and the more positive one loses electrons. This means that silk will lose electrons and cotton will gain electrons.

A material becomes negatively charged when it has an excess of electrons, thus the cotton, which gains electrons, becomes negatively charged.

Note that in this example the numbers are made up to be easy to calculate. In the real world only a tiny fraction of the charges would move from one object to the other, but the total charge would still be conserved.

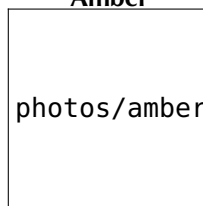
See simulation: (▶) Simulation: VPgsz at www.everythingscience.co.za)

The process of materials becoming charged when they come into contact with other materials is known as tribo-electric charging. Materials can be arranged in a tribo-electric series according to the likelihood of them gaining or losing electrons.

If a material has equal numbers of positive and negative charges we describe it as being neutral (not favouring positive or negative overall charge).

If a neutral material loses electrons it becomes electron deficient and has an overall positive charge. If a neutral material gains electrons it has excess electrons and has an overall negative charge. For this reason we describe the ordering of materials in the tribo-electric series as more positive or more negative depending on whether they are more likely to lose or gain electrons.

Amber



photos/amber_byJohnAlanElson_wikimedia.jpg

Photograph by John Alan Elson on wikimedia

Aluminium

Photograph on wikimedia

This tribo-electric series can allow us to determine whether one material is likely to become charged from another material.

Materials from the more positive end of the series are more likely to lose electrons than those from the more negative end. So when two materials are chosen and rubbed together the one that is more positive in the series will lose electrons and the one that is more negative in the series will gain electrons. For example, amber is more negative than wool and so if a piece of wool is rubbed against a piece of amber then the amber will become negatively charged.

Silicon

Photograph on wikimedia

Material	Tribo-electric series
Glass	Very positive
Human hair	
Nylon	
Wool	
Fur	
Lead	
Silk	
Aluminium	
Paper	
Cotton	
Steel	
Wood	
Amber	
Hard rubber	
Nickel, Copper	
Gold, Platinum	
Polyester	
Polyurethane	
Polypropylene	
Silicon	
Teflon	Very negative

Table 16.1: Tribo-electric series.

Force between charges

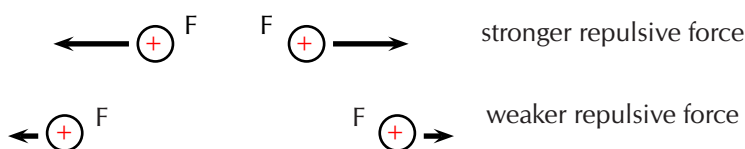
The force exerted by non-moving (static) charges on each other is called the **electrostatic force**. The electrostatic force between:

- **like** charges is **repulsive**
- **opposite** (unlike) charges is **attractive**.

In other words, like charges repel each other while opposite charges attract each other.



The *closer* together the charges are, the *stronger* the electrostatic force between them.



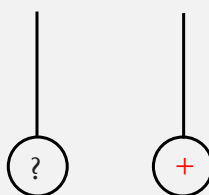
FACT

The word “electron” comes from the Greek word for amber. The ancient Greeks observed that if you rubbed a piece of amber, you could use it to pick up bits of straw.

Example 2: Application of electrostatic forces

QUESTION

Two charged metal spheres hang from strings and are free to move as shown in the picture below. The right hand sphere is positively charged. The charge on the left hand sphere is unknown.



The left sphere is now brought close to the right sphere.

1. If the left hand sphere swings towards the right hand sphere, what can you say about the charge on the left sphere and why?
2. If the left hand sphere swings away from the right hand sphere, what can you say about the charge on the left sphere and why?

SOLUTION

Step 1 : Analyse the problem

In the first case, we have a sphere with positive charge which is *attracting* the left charged sphere. We need to find the charge on the left sphere.

Step 2 : Identify the principles

We are dealing with electrostatic forces between charged objects. Therefore, we know that *like* charges *repel* each other and *opposite* charges *attract* each other.

Step 3 : Apply the principles

- a. In the first case, the positively charged sphere is attracting the left sphere. Since an electrostatic force between unlike charges is attractive, the left sphere must be *negatively* charged.
- b. In the second case, the positively charged sphere repels the left sphere. Like charges repel each other. Therefore, the left sphere must now also be *positively* charged.

Conservation of charge



In all of the examples we've looked at charge was not created or destroyed but it moved from one material to another.

DEFINITION: *Principle of conservation of charge*

The principle of conservation of charge states that the net charge of an isolated system remains constant during any physical process, e.g. two charge objects making contact and separating.

Conductors and insulators



Tip

The effect of the shape on the charge distribution is the reason that we only consider identical conductors for the sharing of charge.

▶ See video: VPfnt at www.everythingscience.co.za

Other materials do not allow the charge carriers, the electrons, to move through them (e.g. plastic, glass). The electrons are bound to the atoms in the material. These materials are called **non-conductors** or **insulators**.

If an excess of charge is placed on an insulator, it will stay where it is put and there will be a concentration of charge in that area of the object. However, if an excess of charge is placed on a conductor, the like charges will repel each other and spread out over the outside surface of the object. When two conductors are made to touch, the total charge on them is shared between the two. If the two conductors are identical, then each conductor will be left with half of the total net charge.

FACT

This collection of charge can actually allow charge to leak off the conductor if the point is sharp enough. It is for this reason that buildings often have a lightning rod on the roof to remove any charge the building has collected. This minimises the possibility of the building being struck by lightning. This “spreading out” of charge would not occur if we were to place the charge on an insulator since charge cannot move in insulators.

Arrangement of charge



The electrostatic force determines the arrangement of charge on the surface of conductors. This is possible because charges can move inside a conductive material. When we place a charge on a spherical conductor the repulsive forces between the individual like charges cause them to spread uniformly over the surface of the sphere. However, for conductors with irregular shapes, there is a concentration of charge near the point or points of the object.

Notice in Figure 16.1 that we show a concentration of charge with more – or + signs, while we represent uniformly spread charges with uniformly spaced – or + signs.

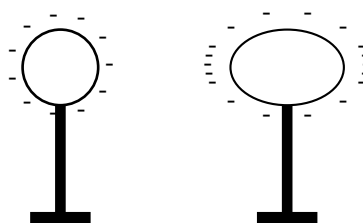


Figure 16.1: Charge on conductors

When two identical conducting spheres on insulating stands are allowed to touch they share the charge evenly between them. If the initial charge on the first sphere is Q_1 and the initial charge on the second sphere is Q_2 , then the final charge on the two spheres after they have been brought into contact is:

$$Q = \frac{Q_1 + Q_2}{2}$$

Quantisation of charge



The basic unit of charge, called the elementary charge, e , is the amount of charge carried by one electron.

Unit of charge



The charge on a single electron is $q_e = 1,6 \times 10^{-19}$ C. All other charges in the universe consist of an integer multiple of this charge. This is known as charge quantisation.

$$Q = nq_e$$

Charge is measured in units called **coulombs (C)**. A coulomb of charge is a very large charge. In electrostatics we therefore often work with charge in micro-coulombs ($1 \mu\text{C} = 1 \times 10^{-6}$ C) and nanocoulombs ($1 \text{nC} = 1 \times 10^{-9}$ C).

Example 3: Charge quantisation

QUESTION

An object has an excess charge of $-1,92 \times 10^{-17}$ C. How many excess electrons

FACT

In 1909 Robert Millikan and Harvey Fletcher measured the charge on an electron. This experiment is now known as Millikan's oil drop experiment. Millikan and Fletcher sprayed oil droplets into the space between two charged plates and used what they knew about forces and in particular the electric force to determine the charge on an electron.

does it have?

SOLUTION

Step 1 : Analyse the problem and identify the principles

We are asked to determine a number of electrons based on a total charge. We know that charge is quantised and that electrons carry the base unit of charge which is $-1,6 \times 10^{-19} \text{ C}$.

Step 2 : Apply the principle

As each electron carries the same charge the total charge must be made up of a certain number of electrons. To determine how many electrons we divide the total charge by the charge on a single electron:

$$\begin{aligned} N &= \frac{-1,92 \times 10^{-17}}{-1,6 \times 10^{-19}} \\ &= 120 \text{ electrons} \end{aligned}$$

Example 4: Conducting spheres and movement of charge

QUESTION

I have 2 charged metal conducting spheres on insulating stands which are identical except for having different charge. Sphere A has a charge of -5 nC and sphere B has a charge of -3 nC . I then bring the spheres together so that they touch each other. Afterwards I move the two spheres apart so that they are no longer touching.

- 1.** *What happens to the charge on the two spheres?*
- 2.** *What is the final charge on each sphere?*

SOLUTION

Step 1 : Analyse the question

We have two identical negatively charged conducting spheres which are brought together to touch each other and then taken apart again.

We need to explain what happens to the charge on each sphere and what the final charge on each sphere is after they are moved apart.

Step 2 : Identify the principles involved

We know that the charge carriers in conductors are free to move around and that charge on a conductor spreads itself out on the surface of the conductor.

Step 3 : Apply the principles

- a. When the two conducting spheres are brought together to touch, it is as though they become one single big conductor and the total charge of the two spheres spreads out across the whole surface of the touching spheres. When the spheres are moved apart again, each one is left with half of the total original charge.
- b. Before the spheres touch, the total charge is: $-5 \text{ nC} + (-3) \text{ nC} = -8 \text{ nC}$. When they touch they share out the -8 nC across their whole surface. When they are removed from each other, each is left with half of the original charge:

$$\frac{-8 \text{ nC}}{2} = -4 \text{ nC}$$

on each sphere.

Example 5: Identical spheres sharing charge I

QUESTION

Two identical, insulated spheres have different charges. Sphere 1 has a charge of $-96 \times 10^{-18} \text{ C}$. Sphere 2 has 60 excess electrons. If the two spheres are brought into contact and then separated, what charge will each have?

SOLUTION

Step 1 : Analyse the question

We need to determine what will happen to the charge when the spheres touch. They are insulators so we know they will NOT allow charge to

move freely. When they touch nothing will happen.

Example 6: Identical spheres sharing charge II

QUESTION

Two identical, metal spheres on insulating stands have different charges. Sphere 1 has a charge of $-9,6 \times 10^{-18} \text{ C}$. Sphere 2 has 60 excess protons. If the two spheres are brought into contact and then separated, what charge will each have? How many electrons or protons does this correspond to?

SOLUTION

Step 1 : Analyse the question

We need to determine what will happen to the charge when the spheres touch. They are metal spheres so we know they will be conductors. This means that the charge is able to move so when they touch it is possible for the charge on each sphere to change. We know that charge will redistribute evenly across the two spheres because of the forces between the charges. We need to know the charge on each sphere, we have been given one.

Step 2 : Identify the principles involved

This problem is similar to the earlier worked example. This time we have to determine the total charge given a certain number of protons. We know that charge is quantised and that protons carry the base unit of charge and are positive so it is $+1,6 \times 10^{-19} \text{ C}$.

Step 3 : Apply the principles

The total charge will therefore be:

$$\begin{aligned} Q_2 &= 60 \times 1,6 \times 10^{-19} \text{ C} \\ &= 9,6 \times 10^{-18} \text{ C} \end{aligned}$$

As the spheres are identical in material, size and shape the charge will redistribute across the two spheres so that it is shared evenly. Each

sphere will have half of the total charge:

$$\begin{aligned}Q &= \frac{Q_1 + Q_2}{2} \\&= \frac{9,6 \times 10^{-18} + (-9,6 \times 10^{-18})}{2} \\&= 0 \text{ C}\end{aligned}$$

So each sphere is now neutral.

No net charge means that there is no excess of electrons or protons.

Example 7: Conservation of charge

QUESTION

Two identical, metal spheres have different charges. Sphere 1 has a charge of $-9,6 \times 10^{-18} \text{ C}$. Sphere 2 has 30 excess electrons. If the two spheres are brought into contact and then separated, what charge will each have? How many electrons does this correspond to?

SOLUTION

Step 1 : Analyse the problem

We need to determine what will happen to the charge when the spheres touch. They are metal spheres so we know they will be conductors. This means that the charge is able to move so when they touch it is possible for the charge on each sphere to change. We know that charge will redistribute evenly across the two spheres because of the forces between the charges. We need to know the charge on each sphere, we have been given one.

Step 2 : Identify the principles

This problem is similar to the earlier worked example. This time we have to determine the total charge given a certain number of electrons. We know that charge is quantised and that electrons carry the base unit of charge which is $-1,6 \times 10^{-19} \text{ C}$. The total charge will therefore be:

$$\begin{aligned}Q_2 &= 30 \times -1,6 \times 10^{-19} \text{ C} \\&= -4,8 \times 10^{-18} \text{ C}\end{aligned}$$

Step 3 : Apply the principles: redistributing charge

As the spheres are identical in material, size and shape the charge will redistribute across the two spheres so that it is shared evenly. Each sphere will have half of the total charge:

$$\begin{aligned} Q &= \frac{Q_1 + Q_2}{2} \\ &= \frac{-9.6 \times 10^{-18} + (-4.8 \times 10^{-18})}{2} \\ &= 7.2 \times 10^{-18} \text{ C} \end{aligned}$$

So each sphere now has:

$$7.2 \times 10^{-18} \text{ C}$$

of charge.

We know that charge is quantised and that electrons carry the base unit of charge which is $-1.6 \times 10^{-19} \text{ C}$.

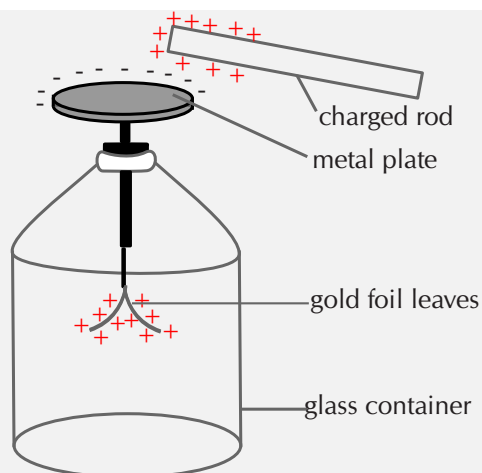
Step 4 : Apply the principles: charge quantisation

As each electron carries the same charge the total charge must be made up of a certain number of electrons. To determine how many electrons we divide the total charge by the charge on a single electron:

$$\begin{aligned} N &= \frac{-7.2 \times 10^{-18}}{-1.6 \times 10^{-19}} \\ &= 45 \text{ electrons} \end{aligned}$$

Investigation: The electroscope

The electroscope is a very sensitive instrument which can be used to detect electric charge. A diagram of a gold leaf electroscope is shown in the figure below. The electroscope consists of a glass container with a metal rod inside which has 2 thin pieces of gold foil attached. The other end of the metal rod has a metal plate attached to it outside the glass container.

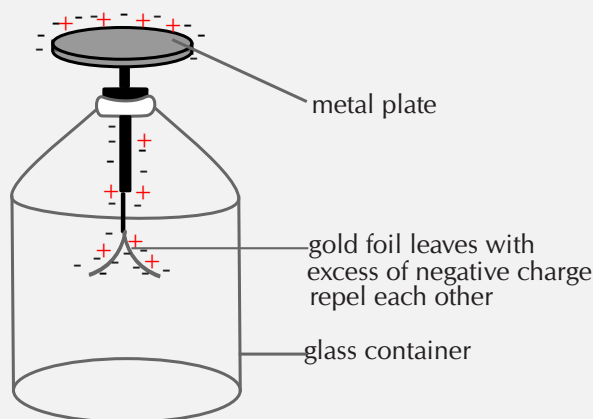


The electroscope detects charge in the following way: A charged object, like the positively charged rod in the picture, is brought close to (but not touching) the neutral metal plate of the electroscope. This causes negative charge in the gold foil, metal rod, and metal plate, to be attracted to the positive rod. Because the metal (gold is a metal too!) is a conductor, the charge can move freely from the foil up the metal rod and onto the metal plate. There is now more negative charge on the plate and more positive charge on the gold foil leaves. This is called *inducing* a charge on the metal plate. It is important to remember that the electroscope is still neutral (the total positive and negative charges are the same), the charges have just been induced to *move* to different parts of the instrument! The induced positive charge on the gold leaves forces them apart since like charges repel! This is how we can tell that the rod is charged. If the rod is now moved away from the metal plate, the charge in the electroscope will spread itself out evenly again and the leaves will fall down because there will no longer be an induced charge on them.

Grounding

If you were to bring the charged rod close to the uncharged electroscope, and then you touched the metal plate with your finger at the same time, this would cause charge to flow up from the ground (the earth), through your body onto the metal plate. Connecting to the earth so charge flows is called **grounding**. The charge flowing onto the plate is opposite to the charge on the rod, since it is attracted to the charge on the rod. Therefore, for our picture, the charge flowing onto the plate would be negative. Now that charge has been added to the electroscope, it is no longer neutral, but has an excess of negative charge. Now if we move the rod away, the leaves will remain apart because they have an excess of negative charge and they repel each other. If we ground

the electroscope again (this time without the charged rod nearby), the excess charge will flow back into the earth, leaving it neutral.

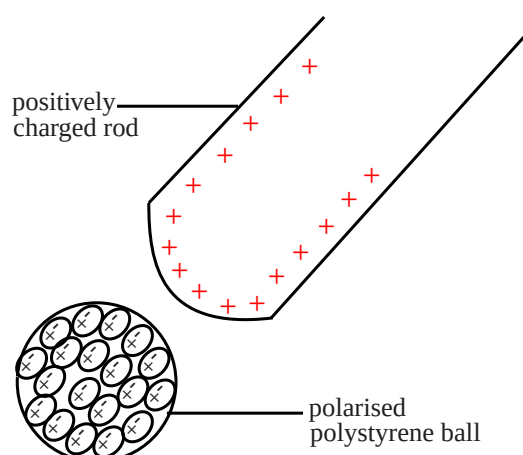


Polarisation



Unlike conductors, the electrons in insulators (non-conductors) are bound to the atoms of the insulator and cannot move around freely through the material. However, a charged object can still exert a force on a neutral insulator due to a phenomenon called **polarisation**.

If a positively charged rod is brought close to a neutral insulator such as polystyrene, it can attract the bound electrons to move round to the side of the atoms which is closest to the rod and cause the positive nuclei to move slightly to the opposite side of the atoms. This process is called *polarisation*. Although it is a very small (microscopic) effect, if there are many atoms and the polarised object is light (e.g. a small polystyrene ball), it can add up to enough force to cause the object to be attracted onto the charged rod. Remember, that the polystyrene is *only* polarised, *not* charged. The polystyrene ball is still neutral since no charge was added or removed from it. The picture shows a not-to-scale view of the polarised atoms in the polystyrene ball:

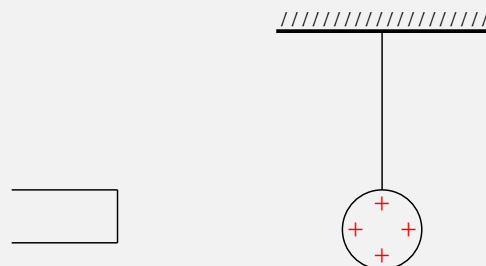


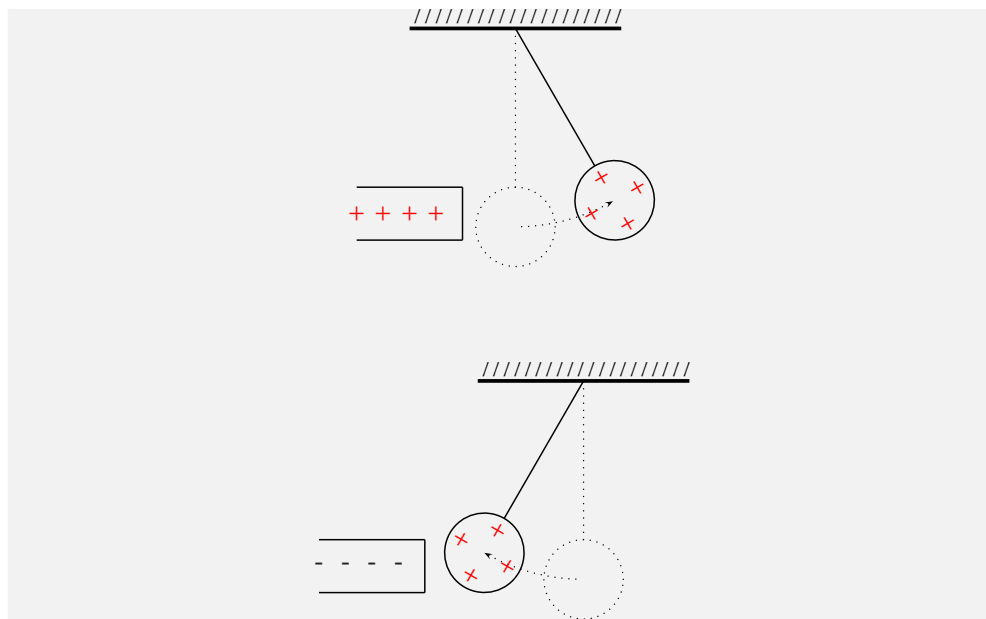
Some materials are made up of molecules which are already polarised. These are molecules which have a more positive and a more negative side but are still neutral overall. Just as a polarised polystyrene ball can be attracted to a charged rod, these materials are also affected if brought close to a charged object.

Investigation: Electrostatic force

You can easily test that like charges repel and unlike charges attract each other by doing a very simple experiment.

Take a glass ball and rub it with a piece of silk, then hang it from its middle with a piece string so that it is free to move. If you then bring another glass rod which you have also charged in the same way next to it, you will see the ball on the string move away from the rod in your hand i.e. it is **repelled**. If, however, you take a plastic rod, rub it with a piece of fur and then bring it close to the ball on the string, you will see the rod on the string move *towards* the rod in your hand i.e. it is **attracted**.





This happens because when you rub the glass with silk, tiny amounts of negative charge are transferred from the glass onto the silk, which causes the glass to have less negative charge than positive charge, making it **positively charged**. When you rub the plastic rod with the fur, you transfer tiny amounts of negative charge onto the rod and so it has more negative charge than positive charge on it, making it **negatively charged**.

Water is an example of a substance which is made of polarised molecules. If a positively charged rod, comb or balloon is brought close to a stream of water, the molecules can rotate so that the negative sides all line up towards the rod. The stream of water will then be attracted to the positively charged object since opposite charges attract.

Water being attracted to a charged balloon



Photograph by NASA

Chapter 16 | Summary

See the summary presentation (📺 Presentation: VPfqd at www.everythingscience.co.za)

- There are two kinds of charge: positive and negative.
- Positive charge is carried by protons in the nucleus.
- Negative charge is carried by electrons.
- Objects can be **positively** charged, **negatively** charged or **neutral**.
- Objects that are neutral have equal numbers of positive and negative charge.
- Unlike charges are attracted to each other and like charges are repelled from each other.
- Charge is neither created nor destroyed, it can only be transferred.
- Charge is measured in coulombs (C).
- Charge is quantised in units of the charge of an electron $1.6 \times 10^{-19} \text{ C}$, $Q = nq_e$.
- Conductors allow charge to move through them easily.
- Insulators do not allow charge to move through them easily.
- Identical, conducting spheres in contact share their charge according to:

$$Q = \frac{Q_1 + Q_2}{2}$$

Physical Quantities		
Quantity	Unit name	Unit symbol
Charge (Q)	coulomb	C
Charge on the electron (q_e)	coulomb	C

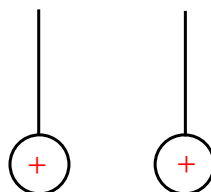
Table 16.2: Units used in **electrostatics**

Chapter 16

End of chapter exercises

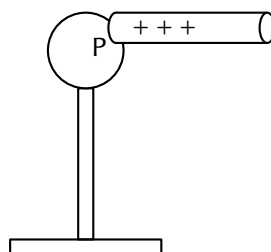
1. What are the two types of charge called?
2. Provide evidence for the existence of two types of charge.
3. Fill in the blanks: The electrostatic force between like charges is _____ while the electrostatic force between opposite charges is _____.
4. I have two positively charged metal balls placed 2 m apart.
 - a. Is the electrostatic force between the balls attractive or repulsive?

- b. If I now move the balls so that they are 1 m apart, what happens to the strength of the electrostatic force between them?
5. I have 2 charged spheres each hanging from string as shown in the picture below.



Choose the correct answer from the options below: The spheres will

- a. swing towards each other due to the attractive electrostatic force between them.
 - b. swing away from each other due to the attractive electrostatic force between them.
 - c. swing towards each other due to the repulsive electrostatic force between them.
 - d. swing away from each other due to the repulsive electrostatic force between them.
6. Describe how objects (insulators) can be charged by contact or rubbing.
7. You are given a perspex ruler and a piece of cloth.
- a. How would you charge the perspex ruler?
 - b. Explain how the ruler becomes charged in terms of charge.
 - c. How does the charged ruler attract small pieces of paper?
8. (IEB 2005/11 HG) An uncharged hollow metal sphere is placed on an insulating stand. A positively charged rod is brought up to touch the hollow metal sphere at P as shown in the diagram below. It is then moved away from the sphere.



Where is the excess charge distributed on the sphere after the rod has been removed?

- a. It is still located at point P where the rod touched the sphere.
- b. It is evenly distributed over the outer surface of the hollow sphere.
- c. It is evenly distributed over the outer and inner surfaces of the hollow sphere.

- d. No charge remains on the hollow sphere.
9. What is the process called where molecules in an uncharged object are caused to align in a particular direction due to an external charge?
10. Explain how an uncharged object can be attracted to a charged object. You should use diagrams to illustrate your answer.
11. Explain how a stream of water can be attracted to a charged rod.
12. An object has an excess charge of $-8,6 \times 10^{-18} \text{ C}$. How many excess electrons does it have?
13. An object has an excess of 235 electrons. What is the charge on the object?
14. An object has an excess of 235 protons. What is the charge on the object?
15. Two identical, metal spheres have different charges. Sphere 1 has a charge of $-4,8 \times 10^{-18} \text{ C}$. Sphere 2 has 60 excess electrons. If the two spheres are brought into contact and then separated, what charge will each have? How many electrons does this correspond to?
16. Two identical, insulated spheres have different charges. Sphere 1 has a charge of $-96 \times 10^{-18} \text{ C}$. Sphere 2 has 60 excess electrons. If the two spheres are brought into contact and then separated, what charge will each have?
17. Two identical, metal spheres have different charges. Sphere 1 has a charge of $-4,8 \times 10^{-18} \text{ C}$. Sphere 2 has 30 excess protons. If the two spheres are brought into contact and then separated, what charge will each have? How many electrons or protons does this correspond to?

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 006b (2.) 006c (3.) 006d (4.) 006e (5.) 006f (6.) 006g
(7.) 006h (8.) 006i (9.) 006j (10.) 006k (11.) 006m (12.) 006n
(13.) 006p (14.) 006q (15.) 006r (16.) 006s (17.) 006t

Potential difference and emf

www ESAFA

When a circuit is connected and complete, charge can move through the circuit. Charge will not move unless there is a reason, a force to drive it round the circuit. Think of it as though charge is at rest and something has to push it along. This means that work needs to be done to make charge move. A force acts on the charges, doing work, to make them move. The force is provided by the battery in the circuit.

A battery has the potential to drive charge round a closed circuit, the battery has potential energy that can be converted into electrical energy by doing work on the charge in the circuit to make it move.

See introductory video: (📺 Video: VPfj at www.everythingscience.co.za)

DEFINITION: Potential Difference

Potential difference is the work done per unit charge, $\frac{W}{q}$. The units of potential difference are the volt (V) which is defined as one joule per coulomb.

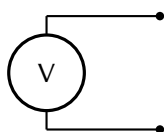
Quantity: Potential difference (V) Unit name: volt Unit symbol: V

Voltmeter

www ESAFB

A voltmeter is an instrument for measuring the potential difference between two points in an electric circuit.

The symbol for a voltmeter is:



A voltmeter



Photography by windy_ on Flickr.com

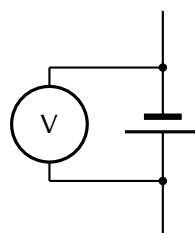
EMF



When you measure the potential difference across (or between) the terminals of a battery that is **not in a complete circuit** you are measuring the emf of the battery. This is the maximum amount of work per coulomb of charge the battery can do to drive charge from one terminal, through the circuit, to the other terminal.

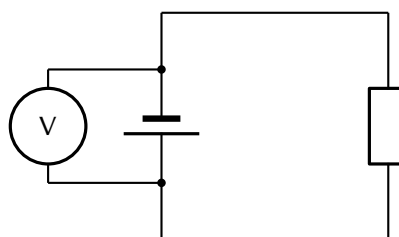
Electrical potential difference is also called voltage.

When you measure the potential difference across (or between) the terminals of a battery that is **in a complete circuit** you are measuring the terminal potential difference of the battery. Although this is measured in volts it is not identical to the emf. The difference will be the work done to drive charge through the battery.



FACT

The volt is named after the Italian physicist Alessandro Volta (1745–1827).



Batteries



Photography on Flickr.com

One lead of the voltmeter is connected to one end of the battery and the other lead is connected to the opposite end. The voltmeter may also be used to measure the voltage across a resistor or any other component of a circuit but must be connected in parallel.

Current

[www](http://www.esafd.com) ESAFD

FACT

Benjamin Franklin made a guess about the direction of charge flow when rubbing smooth wax with rough wool. He thought that the charges flowed from the wax to the wool (i.e. from positive to negative) which was opposite to the real direction. Due to this, electrons are said to have a *negative* charge and so objects which Ben Franklin called “negative” (meaning a shortage of charge) really have an excess of electrons. By the time the true direction of electron flow was discovered, the convention of “positive” and “negative” had already been so well accepted in the scientific world that no effort was made to change it.

Flow of charge

[www](http://www.esafe.com) ESAFE

When we talk about current we talk about how much charge moves past a fixed point in circuit in one second. Think of charges being pushed around the circuit by the battery, there are charges in the wires but unless there is a battery they won't move.

▶ See video: VPfva at www.everythingscience.co.za

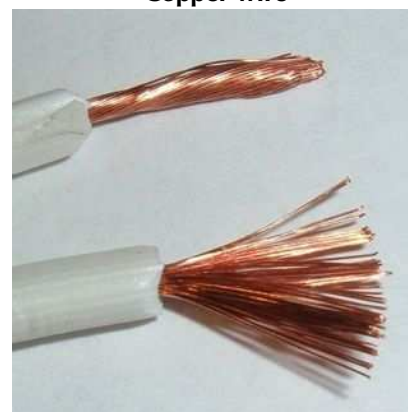
When one charge moves the charges next to it also move. They keep their spacing as if you had a tube of marbles like in this picture or looked at a train and its carriages.



If you push one marble into the tube one must come out the other side, if a train locomotive moves all the carriages move immediately because they are connected. This is similar to charges in the wires of a circuit.

The idea is that if a battery started to drive charge in a circuit all the charges start moving instantaneously.

Copper wire



Photography on Flickr.com

DEFINITION: Current

Current is the rate at which charges moves past a fixed point in a circuit. The units of current are the ampere (A) which is defined as one coulomb per second.

Quantity: Current (I)

Unit name: ampere

Unit symbol: A

Plasma ball



Photography by ahisgett on Flickr.com

We use the symbol I to show current and it is measured in amperes (A). One ampere is one coulomb of charge moving in one second ($\text{C} \cdot \text{s}^{-1}$).

$$I = \frac{Q}{\Delta t}$$

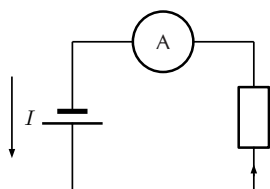
When current flows in a circuit we show this on a diagram by adding arrows. The arrows show the direction of flow in a circuit. By convention we say that charge flows from the positive terminal on a battery to the negative terminal.

If the voltage is high enough a current can be driven through almost anything. In the plasma ball example on the left, a voltage is created that is high enough to get charge to flow through the gas in the ball. The voltage is very high but the resulting current is very low. This makes it safe to touch.

Ammeter

www ESAFF

An ammeter is an instrument used to measure the rate of flow of electric current in a circuit. Since one is interested in measuring the current flowing *through* a circuit component, the ammeter must be connected in *series* with the measured circuit component.



Ammeter

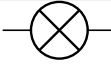
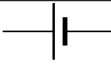
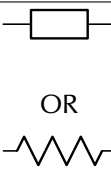
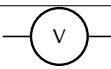
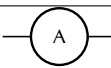


Activity:

Constructing circuits

Construct circuits to measure the emf and the terminal potential difference for a battery. Some common elements (components) which can be found in electrical circuits include light bulbs, batteries, connecting leads, switches, resistors, voltmeters and ammeters. You have learnt about many of these already. Below is a table with

the items and their symbols:

Component	Symbol	Usage
light bulb		glows when charge moves through it
battery		provides energy for charge to move
switch		allows a circuit to be open or closed
resistor	 OR	resists the flow of charge
voltmeter		measures potential difference
ammeter		measures current in a circuit
connecting lead		connects circuit elements together

Tip

A battery **does not** produce the same amount of current no matter what is connected to it. While the voltage produced by a battery is constant, the amount of current supplied depends on what is in the circuit.

Experiment with different combinations of components in the circuits.

The table below summarises the use of each measuring instrument that we discussed and the way it should be connected to a circuit component.

Instrument	Measured Quantity	Proper Connection
Voltmeter	Voltage	In Parallel
Ammeter	Current	In Series

Activity:

Using meters

If possible, connect meters in circuits to get used to the use of meters to measure electrical quantities. If the meters have more than one scale, always connect to the **largest scale** first so that the meter will not be damaged by having to measure values that exceed its limits.

Example 1: Calculating current I**QUESTION**

An amount of charge equal to 45 C moves past a point in a circuit in 1 second, what is the current in the circuit?

SOLUTION**Step 1 : Analyse the question**

We are given an amount of charge and a time and asked to calculate the current. We know that current is the rate at which charge moves past a fixed point in a circuit so we have all the information we need. We have quantities in the correct units already.

Step 2 : Apply the principles

We know that:

$$\begin{aligned} I &= \frac{Q}{\Delta t} \\ I &= \frac{45 \text{ C}}{1 \text{ s}} \\ I &= 45 \text{ C} \cdot \text{s}^{-1} \\ I &= 45 \text{ A} \end{aligned}$$

Step 3 : Quote the final result

The current is 45 A.

Example 2: Calculating current II**QUESTION**

An amount of charge equal to 53 C moves past a fixed point in a circuit in 2

seconds, what is the current in the circuit?

SOLUTION

Step 1 : Analyse the question

We are given an amount of charge and a time and asked to calculate the current. We know that current is the rate at which charge moves past a fixed point in a circuit so we have all the information we need. We have quantities in the correct units already.

Step 2 : Apply the principles

We know that:

$$\begin{aligned} I &= \frac{Q}{\Delta t} \\ I &= \frac{53 \text{ C}}{2 \text{ s}} \\ I &= 26,5 \text{ C} \cdot \text{s}^{-1} \\ I &= 26,5 \text{ A} \end{aligned}$$

Step 3 : Quote the final result

The current is 26,5 A.

Example 3: Calculating current III

QUESTION

95 electrons move past a fixed point in a circuit in one tenth of a second, what is the current in the circuit?

SOLUTION

Step 1 : Analyse the question

We are given a number of charged particles that move past a fixed point and the time that it takes. We know that current is the rate at which charge moves past a fixed point in a circuit so we have to determine the charge. In the last chapter we learnt that the charge carried by an

electron is $1,6 \times 10^{-19} \text{ C}$.

Step 2 : **Apply the principles: determine the charge**

We know that each electron carries a charge of $1,6 \times 10^{-19} \text{ C}$, therefore the total charge is:

$$\begin{aligned} Q &= 95 \times 1,6 \times 10^{-19} \text{ C} \\ &= 1,52 \times 10^{-17} \text{ C} \end{aligned}$$

Step 3 : **Apply the principles: determine the charge**

We know that:

$$\begin{aligned} I &= \frac{Q}{\Delta t} \\ I &= \frac{1,52 \times 10^{-17} \text{ C}}{\frac{1}{10} \text{ s}} \\ I &= \frac{1,52 \times 10^{-17} \text{ C}}{1} \times \frac{1}{\frac{1}{10} \text{ s}} \\ I &= \frac{1,52 \times 10^{-17} \text{ C}}{1} \times \frac{10}{1 \text{ s}} \\ I &= 1,52 \times 10^{-16} \text{ C} \cdot \text{s}^{-1} \\ I &= 1,52 \times 10^{-16} \text{ A} \end{aligned}$$

Step 4 : **Quote the final result**

The current is $1,52 \times 10^{-16} \text{ A}$.

Resistance



Resistance is a measure of "how hard" it is to "push" electricity through a circuit element.

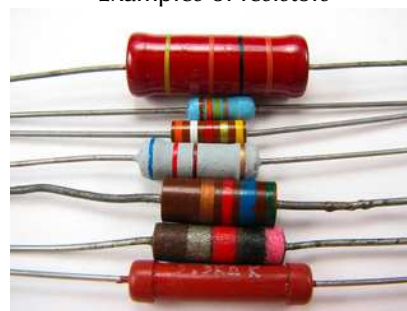
Resistance can also apply to an entire circuit. ▶ See video: VPfvk at www.everythingscience.co.za

What causes resistance?



On a microscopic level, electrons moving through the conductor collide (or interact) with the particles of which the conductor (metal) is made. When they collide, they transfer kinetic energy. The electrons therefore lose kinetic energy and slow down. This leads to resistance. The transferred energy causes the resistor to heat up. You can feel this directly if you touch a cellphone charger when you are charging a cell phone - the charger gets warm because its circuits have some resistors in them!

Examples of resistors



Photograph by oskay on Flickr.com

DEFINITION: Resistance

Resistance slows down the flow of charge in a circuit. The unit of resistance is the ohm (Ω) which is defined as a volt per ampere of current.

Quantity: Resistance R Unit: ohm Unit symbol: ω

FACT

Fluorescent lightbulbs do not use thin wires; they use the fact that certain gases glow when a current flows through them. They are much more efficient (much less resistance) than lightbulbs.

$$1 \text{ ohm} = 1 \frac{\text{volt}}{\text{ampere}}$$

Light bulb filament



Photograph by clagnut on Flickr.com

All conductors have some resistance. For example, a piece of wire has less resistance than a light bulb, but both have resistance.

A lightbulb is a very thin wire surrounded by a glass housing. The high resistance of the small wire (filament) in a lightbulb causes the electrons to transfer a lot of their kinetic energy in the form of heat. The heat energy is enough to cause the filament to glow white-hot which produces light.

The wires connecting the lamp to the cell or battery hardly even get warm while conducting the same amount of current. This is because of their much lower resistance due to their larger cross-section (they are thicker).

An important effect of a resistor is that it *converts* electrical energy into other forms of **heat** energy. **Light** energy is a by-product of the heat that is produced.

Physical attributes affecting resistance [NOT IN CAPS]

The physical attributes of a resistor affect its total resistance.

- **Length:** if a resistor is increased in length its resistance will increase. Typically if you increase the length of a resistor by a certain factor you will increase the resistance by the same factor.
- **Width and height or cross-sectional area:** if a resistor provides a larger pathway by being made wider or broader then more current can flow through it. If the total surface area through which current flows (cross-sectional area) is increased by a factor the resistance typically decreases by the same factor.

Extension: For a single resistor this can be summarised as

$$R \propto \frac{L}{A},$$

where L is the length and A is the cross-sectional area.

Why do batteries go flat?

A battery stores chemical potential energy. When it is connected in a circuit, a chemical reaction takes place inside the battery which converts chemical potential energy to electrical energy which powers the electrons to move through the circuit. All the circuit elements (such as the conducting leads, resistors and lightbulbs) have some resistance to the flow of charge and convert the electrical energy to heat and, in the case of the lightbulb, light. Since energy is always conserved, the battery goes flat when all its chemical potential energy has been converted into other forms of energy.

FACT

There is a special type of conductor, called a **superconductor** that has no resistance, but the materials that make up all known superconductors only start superconducting at very low temperatures. The “highest” temperature superconductor is mercury barium calcium copper oxide ($\text{HgBa}_2\text{Ca}_2\text{Cu}_3\text{O}_x$) which is superconducting for temperatures of -140°C and colder.

Resistors in electric circuits



It is important to understand what effect adding resistors to a circuit has on the *total* resistance of a circuit and on the current that can flow in the circuit.

Series resistors

When we add resistors in series to a circuit:

- There is only one path for current to flow which ensures that the **current is the same at every point in the circuit**.
- The voltage is **divided** across the resistors. The voltage across the battery in the circuit is equal to the sum of voltages across the series resistors:

$$V_{\text{battery}} = V_1 + V_2 + \dots$$

- The resistance to the flow of current **increases**. The total resistance, R_S is given by:

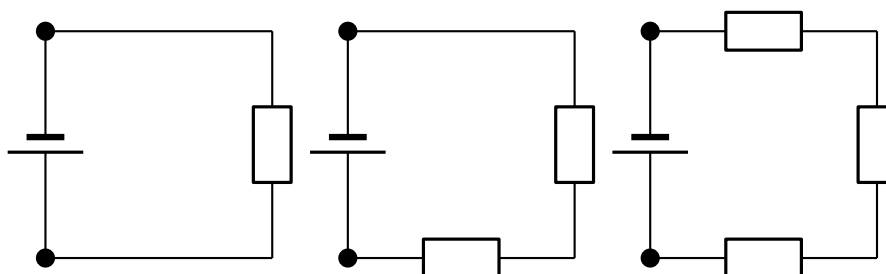
$$R_S = R_1 + R_2 + \dots$$

We will revisit each of these features of series circuits in more detail below.

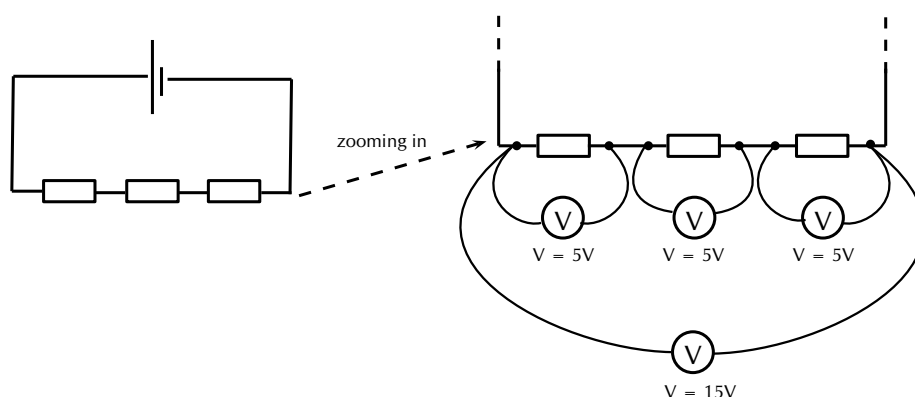
When resistors are in series, one after the other, there is a potential difference across each resistor. The total potential difference across a set of resistors in series is the sum of the potential differences across each of the resistors in the set.

$$V_{\text{battery}} = V_1 + V_2 + \dots$$

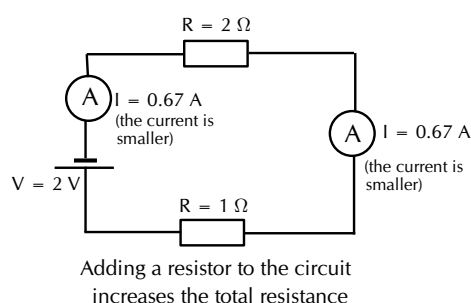
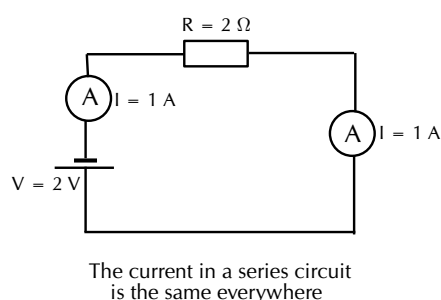
Look at the circuits below. If we measured the potential difference between the black dots in all of these circuits it would be the same as we saw earlier. So we now know the total potential difference is the same across one, two or three resistors. We also know that some work is required to make charge flow through each resistor.



Let us look at this in a bit more detail. In the picture below you can see what the different measurements for 3 identical resistors in series could look like. The total voltage across all three resistors is the sum of the voltages across the individual resistors. Resistors in series are known as voltage dividers because the total voltage across all the resistors is divided amongst the individual resistors.



Consider the diagram below. On the left there is a circuit with a single resistor and a battery. **No matter where we measure the current, it is the same in a series circuit.** On the right, we have added a second resistor in series to the circuit. The *total* resistance of the circuit has *increased* and you can see from the reading on the ammeter that the current in the circuit has decreased.



General experiment: Voltage dividers

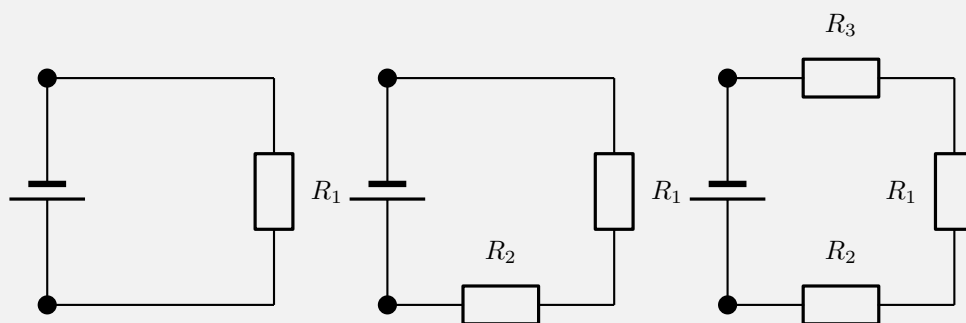
Aim: Test what happens to the current and the voltage in series circuits when additional resistors are added.

Apparatus:

- A battery
- A voltmeter
- An ammeter
- Wires
- Resistors

Method:

- Construct each circuit shown below
- Measure the voltage across each resistor in the circuit.
- Measure the current before and after each resistor in the circuit.

**Results and conclusions:**

- Compare the sum of the voltages across all the resistors in each of the circuits.
- Compare the various current measurements within the same circuit.

The total resistance is equal to R_1 in the first circuit, to $R_1 + R_2$ in the second circuit and $R_1 + R_2 + R_3$ in the third circuit. In general, for resistors in series, the total resistance is given by:

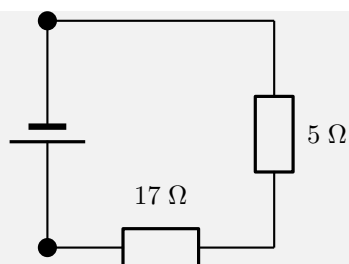
$$R_S = R_1 + R_2 + \dots$$

We know that the potential energy lost across a resistor is proportional to the resistance of the component because the higher the resistance the more work must be done to drive charge through the resistor.

Example 4: Series resistors I

QUESTION

A circuit contains two resistors in series. The resistors have resistance values of $5\ \Omega$ and $17\ \Omega$.



What is the total resistance in the circuit?

SOLUTION

Step 1 : Analyse the question

We are told that the circuit is a series circuit and that we need to calculate the total resistance. The values of the two resistors have been given in the correct units, Ω .

Step 2 : Apply the relevant principles

The total resistance for resistors in series is the sum of the individual resistances. We can use

$$R_S = R_1 + R_2 + \dots$$

. We have only two resistors and we now the resistances. In this case we have that:

$$R_S = R_1 + R_2 + \dots$$

$$R_S = R_1 + R_2$$

$$= 5 \, \Omega + 17 \, \Omega$$

$$= 22 \, \Omega$$

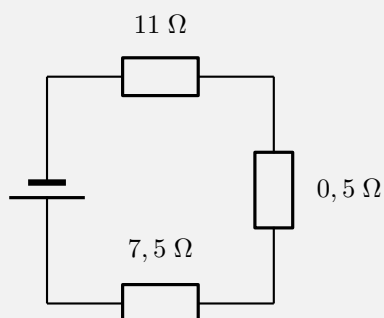
Step 3 : Quote the final result

The total resistance of the resistors in series is $22 \, \Omega$.

Example 5: Series resistors II

QUESTION

A circuit contains three resistors in series. The resistors have resistance values of $0,5\ \Omega$, $7,5\ \Omega$ and $11\ \Omega$.



What is the total resistance in the circuit?

SOLUTION

Step 1 : Analyse the question

We are told that the circuit is a series circuit and that we need to calculate the total resistance. The values of the three resistors have been given in the correct units, Ω .

Step 2 : Apply the relevant principles

The total resistance for resistors in series is the sum of the individual resistances. We can use

$$R_S = R_1 + R_2 + \dots$$

. We have three resistors and we now the resistances. In this case we have that:

$$\begin{aligned} R_S &= R_1 + R_2 + \dots \\ R_S &= R_1 + R_2 + R_3 \\ &= 0,5\ \Omega + 7,5\ \Omega + 11\ \Omega \\ &= 19\ \Omega \end{aligned}$$

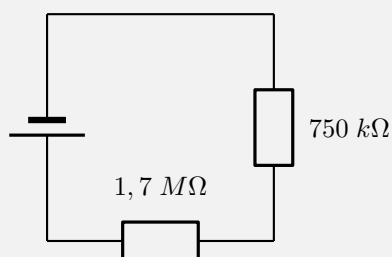
Step 3 : Quote the final result

The total resistance of the resistors in series is $19\ \Omega$.

Example 6: Series resistors III

QUESTION

A circuit contains two resistors in series. The resistors have resistance values of $750\ k\Omega$ and $1,7\ M\Omega$.



What is the total resistance in the circuit?

SOLUTION

Step 1 : Analyse the question

We are told that the circuit is a series circuit and that we need to calculate the total resistance. The values of the two resistors have been given in the correct units, Ω .

Step 2 : Apply the relevant principles

The total resistance for resistors in series is the sum of the individual resistances. We can use

$$R_S = R_1 + R_2 + \dots$$

. We have only two resistors and we know the resistances. In this case

we have that:

$$R_S = R_1 + R_2 + \dots$$

$$R_S = R_1 + R_2$$

$$= 750 \text{ k}\Omega + 1,7 \text{ M}\Omega$$

$$= 750 \times 10^3 \Omega + 1,7 \times 10^6 \Omega$$

$$= 0,75 \times 10^6 \Omega + 1,7 \times 10^6 \Omega$$

$$= 2,45 \times 10^6 \Omega = 2,45 \text{ M}\Omega$$

Step 3 : **Quote the final result**

The total resistance of the resistors in series is 2,45 MΩ.

Parallel resistors



When we add resistors in parallel to a circuit:

- There are more paths for current to flow which ensures that the **current splits across the different paths**.
- The voltage is **the same** across the resistors. The voltage across the battery in the circuit is equal to the voltage across each of the parallel resistors:

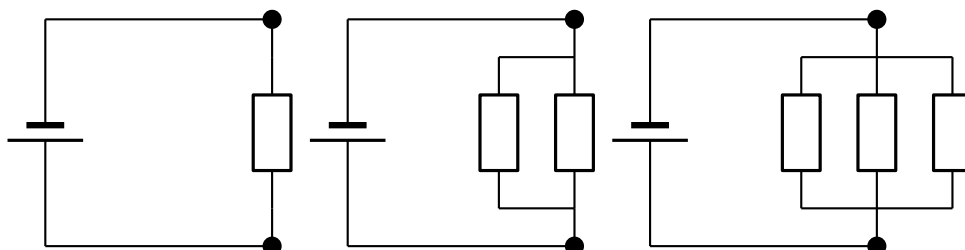
$$V_{\text{battery}} = V_1 = V_2 = V_3 \dots$$

- The resistance to the flow of current **decreases**. The total resistance, R_P is given by:

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

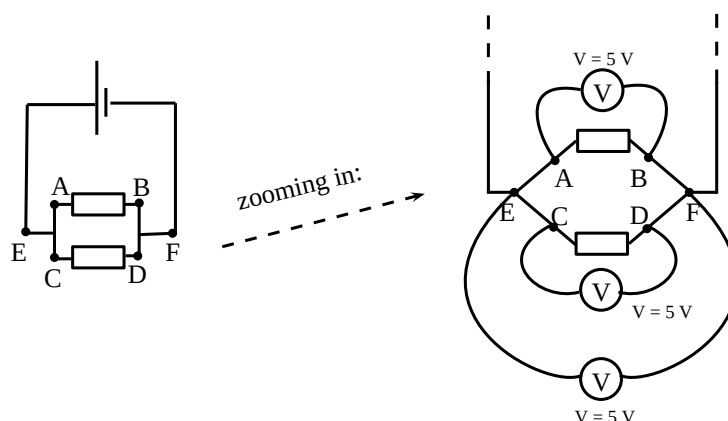
When resistors are connected in parallel the start and end points for all the resistors are the same. These points have the same potential energy and so the potential difference between them is the same no matter what is put in between them. You can have one, two or many resistors between the two points, the potential difference will not change. You can ignore whatever components are between two points in a circuit when calculating the difference between the two points.

Look at the following circuit diagrams. The battery is the same in all cases, all that changes is more resistors are added between the points marked by the black dots. If we were to measure the potential difference between the two dots in these circuits we would get the same answer for all three cases.



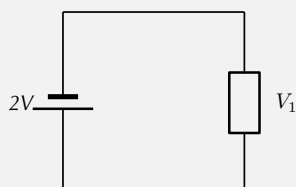
Lets look at two resistors in parallel more closely. When you construct a circuit you use wires and you might think that measuring the voltage in different places on the wires will make a difference. This is not true. The potential difference or voltage measurement will only be different if you measure a different set of components. All points on the wires that have no circuit components between them will give you the same measurements.

All three of the measurements shown in the picture below (i.e. A–B, C–D and E–F) will give you the same voltage. The different measurement points on the left have no components between them so there is no change in potential energy. Exactly the same applies to the different points on the right. When you measure the potential difference between the points on the left and right you will get the same answer.



Example 7: Voltages I**QUESTION**

Consider this circuit diagram:



What is the voltage across the resistor in the circuit shown?

SOLUTION**Step 1 : Check what you have and the units**

We have a circuit with a battery and one resistor. We know the voltage across the battery. We want to find that voltage across the resistor.

$$V_{\text{battery}} = 2 \text{ V}$$

Step 2 : Applicable principles

We know that the voltage across the battery must be equal to the total voltage across all other circuit components.

$$V_{\text{battery}} = V_{\text{total}}$$

There is only one other circuit component, the resistor.

$$V_{\text{total}} = V_1$$

This means that the voltage across the battery is the same as the voltage across the resistor.

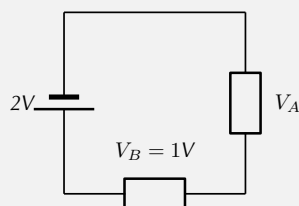
$$V_{\text{battery}} = V_{\text{total}} = V_1$$

$$V_{\text{battery}} = V_{\text{total}} = V_1$$

$$V_1 = 2V$$

Example 8: Voltages II**QUESTION**

Consider this circuit:



What is the voltage across the unknown resistor in the circuit shown?

SOLUTION**Step 1 : Check what you have and the units**

We have a circuit with a battery and two resistors. We know the voltage across the battery and one of the resistors. We want to find that voltage across the resistor.

$$V_{\text{battery}} = 2V$$

$$V_B = 1V$$

Step 2 : Applicable principles

We know that the voltage across the battery must be equal to the total voltage across all other circuit components that are in series.

$$V_{\text{battery}} = V_{\text{total}}$$

The total voltage in the circuit is the sum of the voltages across the individual resistors

$$V_{\text{total}} = V_A + V_B$$

Using the relationship between the voltage across the battery and total voltage across the resistors

$$V_{\text{battery}} = V_{\text{total}}$$

$$V_{\text{battery}} = V_1 + V_{\text{resistor}}$$

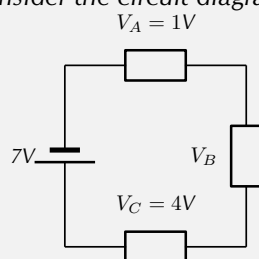
$$2V = V_1 + 1V$$

$$V_1 = 1V$$

Example 9: Voltages III

QUESTION

Consider the circuit diagram:



What is the voltage across the unknown resistor in the circuit shown?

SOLUTION

Step 1 : **Check what you have and the units**

We have a circuit with a battery and three resistors. We know the voltage across the battery and two of the resistors. We want to find that voltage across the unknown resistor.

$$V_{\text{battery}} = 7V$$

$$V_{\text{known}} = V_A + V_C$$

$$= 1V + 4V$$

Step 2 : **Applicable principles**

We know that the voltage across the battery must be equal to the total voltage across all other circuit components that are in series.

$$V_{\text{battery}} = V_{\text{total}}$$

The total voltage in the circuit is the sum of the voltages across the

individual resistors

$$V_{total} = V_B + V_{known}$$

Using the relationship between the voltage across the battery and total voltage across the resistors

$$V_{battery} = V_{total}$$

$$V_{battery} = V_B + V_{known}$$

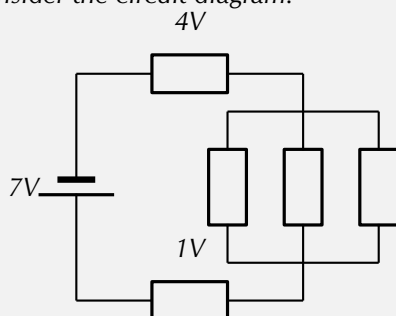
$$7V = V_B + 5V$$

$$V_B = 2V$$

Example 10: Voltages IV

QUESTION

Consider the circuit diagram:



What is the voltage across the parallel resistor combination in the circuit shown?

Hint: the rest of the circuit is the same as the previous problem.

SOLUTION

Step 1 : Quick Answer

The circuit is the same as the previous example and we know that the voltage difference between two points in a circuit does not depend on what is between them so the answer is the same as above $V_{parallel} = 2V$.

Step 2 : Check what you have and the units - long answer

We have a circuit with a battery and five resistors (two in series and

three in parallel). We know the voltage across the battery and two of the resistors. We want to find that voltage across the parallel resistors, $V_{parallel}$.

$$V_{battery} = 7V$$

$$V_{known} = 1V + 4V$$

Step 3 : Applicable principles

We know that the voltage across the battery must be equal to the total voltage across all other circuit components.

$$V_{battery} = V_{total}$$

Voltages only add algebraically for components in series. The resistors in parallel can be thought of as a single component which is in series with the other components and then the voltages can be added.

$$V_{total} = V_{parallel} + V_{known}$$

Using the relationship between the voltage across the battery and total voltage across the resistors

$$V_{battery} = V_{total}$$

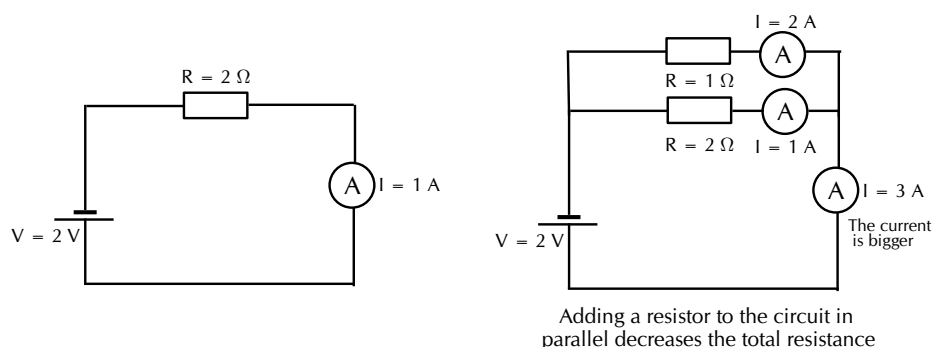
$$V_{battery} = V_{parallel} + V_{known}$$

$$7V = V_{parallel} + 5V$$

$$V_{parallel} = 2V$$

In contrast to the series case, when we add resistors in parallel, we create **more paths** along which current can flow. By doing this we *decrease* the total resistance of the circuit!

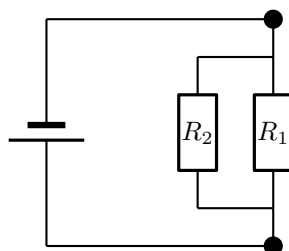
Take a look at the diagram below. On the left we have the same circuit as in the previous section with a battery and a resistor. The ammeter shows a current of 1 A. On the right we have added a second resistor in parallel to the first resistor. This has increased the number of paths (branches) the charge can take through the circuit - the total resistance has decreased. You can see that the current in the circuit has increased. Also notice that the current in the different branches can be different.



The total resistance of a number of parallel resistors is NOT the sum of the individual resistances as the overall resistance decreases with more paths for the current. The total resistance for parallel resistors is given by:

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

Let us consider the case where we have two resistors in parallel and work out what the final resistance would be. This situation is shown in the diagram below:



Applying the formula for the total resistance we have:

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

There are only two resistors

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2}$$

Add the fractions

$$\frac{1}{R_P} = \frac{1}{R_1} \times \frac{R_2}{R_2} + \frac{1}{R_2} \times \frac{R_1}{R_1}$$

$$\frac{1}{R_P} = \frac{R_2}{R_1 R_2} + \frac{R_1}{R_1 R_2}$$

$$\begin{aligned} & \text{Rearrange} \\ \frac{1}{R_P} &= \frac{R_2 + R_1}{R_1 R_2} \\ \frac{1}{R_P} &= \frac{R_1 + R_2}{R_1 R_2} \\ R_P &= \frac{R_1 R_2}{R_1 + R_2} \end{aligned}$$

For any two resistors in parallel we now know that

$$R_P = \frac{\text{product of resistances}}{\text{sum of resistances}} = \frac{R_1 R_2}{R_1 + R_2}$$

General experiment: Current dividers

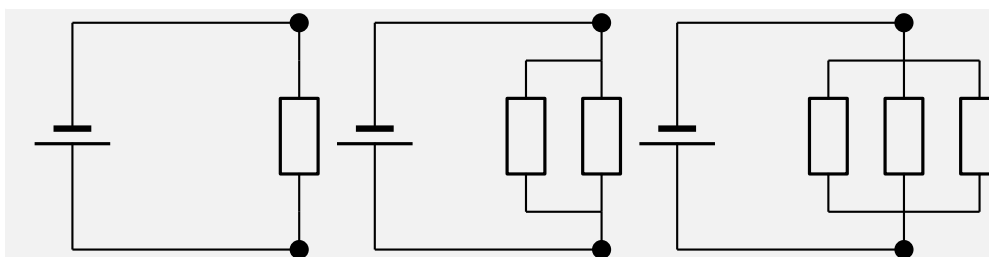
Aim: Test what happens to the current and the voltage in series circuits when additional resistors are added.

Apparatus:

- A battery
- A voltmeter
- An ammeter
- Wires
- Resistors

Method:

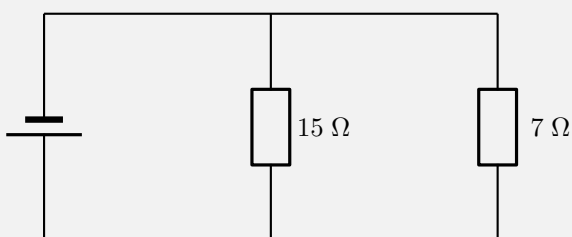
- Connect each circuit shown below
- Measure the voltage across each resistor in the circuit.
- Measure the current before and after each resistor in the circuit and before and after the parallel branches.

**Results and conclusions:**

- Compare the currents through individual resistors with each other.
- Compare the sum of the currents through individual resistors with the current before the parallel branches.
- Compare the various voltage measurements across the parallel resistors.

Example 11: Parallel resistors I**QUESTION**

A circuit contains two resistors in parallel. The resistors have resistance values of $15\ \Omega$ and $7\ \Omega$.



What is the total resistance in the circuit?

SOLUTION**Step 1 : Analyse the question**

We are told that the resistors in the circuit are in parallel circuit and that we need to calculate the total resistance. The values of the two resistors have been given in the correct units, Ω .

Step 2 : Apply the relevant principles

The total resistance for resistors in parallel has been shown to be the

product of the resistances divided by the sum. We can use

$$R_P = \frac{R_1 R_2}{R_1 + R_2}$$

. We have only two resistors and we now the resistances. In this case we have that:

$$\begin{aligned} R_P &= \frac{R_1 R_2}{R_1 + R_2} \\ &= \frac{(15 \, \Omega)(7 \, \Omega)}{15 \, \Omega + 7 \, \Omega} \\ &= \frac{105 \, \Omega^2}{22 \, \Omega} \\ &= 4,77 \, \Omega \end{aligned}$$

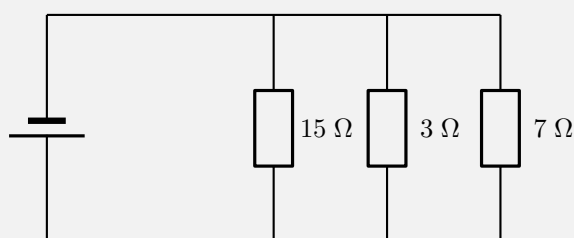
Step 3 : Quote the final result

The total resistance of the resistors in parallel is 4,77 Ω .

Example 12: Parallel resistors II

QUESTION

We add a third parallel resistor to the configuration (setup) in the previous example. The additional resistor has a resistance of .



What is the total resistance in the circuit?

SOLUTION

Step 1 : Analyse the question

We are told that the resistors in the circuit are in parallel circuit and that we need to calculate the total resistance. The value of the additional resistor has been given in the correct units, Ω .

Step 2 : Apply the relevant principles

The total resistance for resistors in parallel has been given as

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

. We have three resistors and we now the resistances. In this case we have that:

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

there are three resistors

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

add the fractions

$$\begin{aligned}\frac{1}{R_P} &= \frac{1}{R_1} \times \frac{R_2 R_3}{R_2 R_3} + \frac{1}{R_2} \times \frac{R_1 R_3}{R_1 R_3} + \frac{1}{R_3} \times \frac{R_1 R_2}{R_1 R_2} \\ \frac{1}{R_P} &= \frac{R_2 R_3}{R_1 R_2 R_3} + \frac{R_1 R_3}{R_1 R_2 R_3} + \frac{R_1 R_2}{R_1 R_2 R_3}\end{aligned}$$

rearrange

$$\begin{aligned}\frac{1}{R_P} &= \frac{R_2 R_3 + R_1 R_3 + R_1 R_2}{R_1 R_2 R_3} \\ R_P &= \frac{R_1 R_2 R_3}{R_2 R_3 + R_1 R_3 + R_1 R_2} \\ R_P &= \frac{(15 \Omega)(7 \Omega)(3 \Omega)}{(7 \Omega)(3 \Omega) + (15 \Omega)(3 \Omega) + (7 \Omega)(15 \Omega)} \\ R_P &= \frac{315 \Omega^3}{21 \Omega^2 + 45 \Omega^2 + 105 \Omega^2} \\ R_P &= \frac{315 \Omega^3}{171 \Omega^2} \\ R_P &= 1,84 \Omega\end{aligned}$$

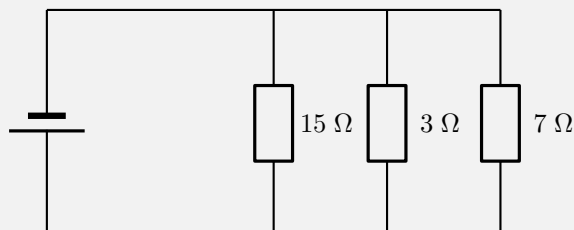
Step 3 : Quote the final result

The total resistance of the resistors in parallel is $1,84 \Omega$. It makes sense that this is less than in the previous example because we added an additional path and reduced the overall resistance in the circuit.

When calculating the resistance for complex resistor configurations, you can start with any combination of two resistors (in series or parallel) and calculate their total resistance. Then you can replace them with a single resistor that has the total resistance you calculated. Now use this new resistor in combination with any other resistor and repeat the process until there is only one resistor left. In the above example we could just have used the answer from the first example in parallel with the new resistor and we would get the same answer.

Example 13: Parallel resistors III**QUESTION**

We add a third parallel resistor to the first parallel worked example configuration (setup). The additional resistor has a resistance of $3\ \Omega$.



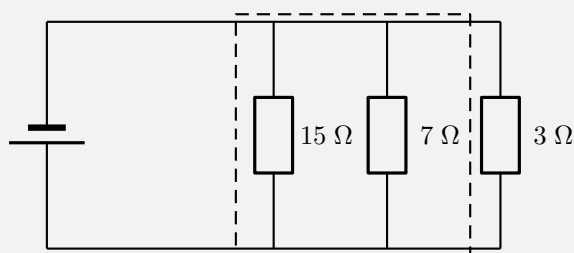
What is the total resistance in the circuit?

SOLUTION**Step 1 : Analyse the question**

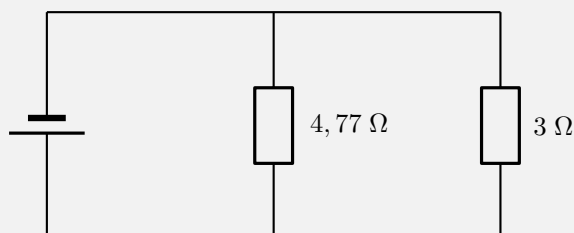
We are told that the resistors in the circuit are in parallel circuit and that we need to calculate the total resistance. The value of the additional resistor has been given in the correct units, Ω .

Step 2 : Apply the relevant principles

We can swap the resistors around without changing the circuit:



We have already calculated the total resistance of the two resistors in the dashed box to be $4,77\ \Omega$. We can replace these two resistors with a single resistor of $4,77\ \Omega$ to get:



Step 3 : Calculate the total resistance for the next pair of resistors

Then we use the formula for two parallel resistors again to get the total resistance for this new circuit:

$$\begin{aligned}R_P &= \frac{R_1 R_2}{R_1 + R_2} \\&= \frac{(4,77 \, \Omega)(3 \, \Omega)}{4,77 \, \Omega + 3 \, \Omega} \\&= \frac{14,31 \, \Omega^2}{11,77 \, \Omega} \\&= 1,84 \, \Omega\end{aligned}$$

Step 4 : Quote the final result

The total resistance of the resistors in parallel is 1,84 Ω . This is the same result as when we added all three resistors together at once.

Exercise 17 - 1

1. What is the unit of resistance called and what is its symbol?
2. Explain what happens to the total resistance of a circuit when resistors are added in series?
3. Explain what happens to the total resistance of a circuit when resistors are added in parallel?
4. Why do batteries go flat?

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 006u (2.) 006v (3.) 006w (4.) 006x

See simulation: () Simulation: VPfzj at www.everythingscience.co.za

Chapter 17 | Summary

See the summary presentation (📺 Presentation: VPftf at www.everythingscience.co.za)

- The potential difference across the terminals of a battery when it is **not** in a complete circuit is the electromotive force (emf) measured in volts (V).
- The potential difference across the terminals of a battery when it is in a complete circuit is the terminal potential difference measured in volts (V).
- Voltage is a measure of required/done to move a certain amount of charge and is equivalent to $\text{J} \cdot \text{C}^{-1}$.
- Current is the rate at which charge moves/flows and is measured in amperes (A) which is equivalent to $\text{C} \cdot \text{s}^{-1}$.
- Conventional current flows from the positive terminal of a battery, through a circuit, to the negative terminal.
- Ammeters measure current and must be connected in series.
- Voltmeters measure potential difference (voltage) and must be connected in parallel.
- Resistance is a measure of how much work must be done for charge to flow through a circuit element and is measured in ohms (Ω) and is equivalent to $\text{V} \cdot \text{A}^{-1}$.
- Resistance of circuit elements is related to the material from which they are made as well as the physical characteristics of length and cross-sectional area.
- Current is constant through resistors in series and they are called voltage dividers as the sum of the voltages is equal to the voltage across the entire set of resistors.
- The total resistance of resistors in series is the sum of the individual resistances, $R_S = R_1 + R_2 + \dots$
- Voltage is constant across resistors in parallel and they are called current divides because the sum of the current through each is the same as the total current through the circuit configuration.
- The total resistance of resistors in parallel is calculated by using $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$ which is $R_P = \frac{R_1 R_2}{R_1 + R_2}$ for two parallel resistors.

Physical Quantities		
Quantity	Unit name	Unit symbol
Potential difference (V)	volt	V
emf	volt	V
Voltage (V)	volt	V
Current (I)	ampere	A
Resistance (R)	ohm	Ω

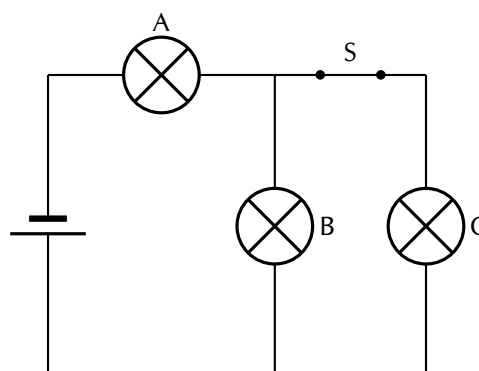
Table 17.1: Units used in **electric circuits**

Chapter 17**End of chapter exercises**

1. Write definitions for each of the following:
 - a. resistor
 - b. coulomb
 - c. voltmeter
2. Draw a circuit diagram which consists of the following components:
 - a. 2 batteries in parallel
 - b. an open switch
 - c. 2 resistors in parallel
 - d. an ammeter measuring total current
 - e. a voltmeter measuring potential difference across one of the parallel resistors
3. Complete the table below:

Quantity	Symbol	Unit of measurement	Symbol of unit
e.g. Distance	e.g. D	e.g. kilometre	e.g. km
Resistance			
Current			
Potential difference			

4. [SC 2003/11] The emf of a battery can best be explained as the ...
 - a. rate of energy delivered per unit current
 - b. rate at which charge is delivered
 - c. rate at which energy is delivered
 - d. charge per unit of energy delivered by the battery
5. [IEB 2002/11 HG1] Which of the following is the correct definition of the emf of a battery?
 - a. It is the product of current and the external resistance of the circuit.
 - b. It is a measure of the cell's ability to conduct an electric current.
 - c. It is equal to the "lost volts" in the internal resistance of the circuit.
 - d. It is the power supplied by the battery per unit current passing through the battery.
6. [IEB 2005/11 HG] Three identical light bulbs A, B and C are connected in an electric circuit as shown in the diagram below.



- a. How bright is bulb A compared to B and C?
- b. How bright are the bulbs after switch S has been opened?
- c. How do the currents in bulbs A and B change when switch S is opened?

	Current in A	Current in B
(a)	decreases	increases
(b)	decreases	decreases
(c)	increases	increases
(d)	increases	decreases

7. [IEB 2004/11 HG1] When a current I is maintained in a conductor for a time of t , how many electrons with charge e pass any cross-section of the conductor per second?
 - a. It
 - b. It/e
 - c. Ite
 - d. e/It
8. If you have a circuit consisting of 4 resistors of equal resistance in series and the total voltage across all of them is 17 V, what is the voltage across each of them?
9. If you have a circuit consisting of 4 resistors of equal resistance in parallel and the total voltage across all of them is 17 V, what is the voltage across each of them?
10. In a circuit consisting of a battery and 3 resistors in series, what is the voltage across the first resistor if the voltage across the battery is 12 V and the voltages across the other two resistors is 3 V and 2 V respectively?
11. There are 3 resistors in parallel with resistances of $3\ \Omega$, $4\ \Omega$ and $11\ \Omega$. What is the total resistance of the parallel combination?
12. The same three resistors as above are now arranged in series, $3\ \Omega$, $4\ \Omega$ and $11\ \Omega$. What is the total resistance of the series combination?
13. The total resistance of two resistors in parallel is $3\ \Omega$, the one resistor has a resistance of $5\ \Omega$, what is the resistance of the other resistor?
14. In a series circuit there are 3 resistors with voltages of 2 V, 5 V and 8 V,

what is the voltage across the battery in the circuit?

15. In a parallel circuit there are 3 resistors with voltages of 2 V, 2 V and 2 V,
what is the voltage across the battery in the circuit?

 More practice  video solutions  or help at www.everythingscience.co.za

- (1.) 006y (2.) 006z (3.) 0070 (4.) 0071 (5.) 0072 (6.) 0073
(7.) 0074 (8.) 01uw (9.) 01ux (10.) 01uy (11.) 01uz (12.) 01v0
(13.) 01v1 (14.) 01v2 (15.) 01v3

Reactions in aqueous solution

18

Introduction

 ESAFL

Many reactions in chemistry and all biological reactions (reactions in living systems) take place in water. We say that these reactions take place in aqueous solution. Water has many unique properties and is plentiful on Earth. For these reasons reactions in aqueous solutions occur frequently. In this chapter we will look at some of these reactions in detail. Almost all the reactions that occur in aqueous solutions involve ions. We will look at three main types of reactions that occur in aqueous solutions, namely precipitation reactions, acid-base reactions and redox reactions. Before we can learn about the types of reactions, we need to first look at ions in aqueous solutions and electrical conductivity. See introductory video: (📺 Video: VPbls at www.everythingscience.co.za)

Ions in aqueous solution

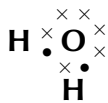
 ESAFM

Water is seldom pure. Because of the structure of the water molecule, substances can dissolve easily in it. This is very important because if water wasn't able to do this, life would not be possible on Earth. In rivers and the oceans for example, dissolved oxygen means that organisms (such as fish) are able to respire (breathe). For plants, dissolved nutrients are available in a form which they can absorb. In the human body, water is able to carry dissolved substances from one part of the body to another.

Dissociation in water

 ESAFN

Water is a **polar molecule**. If we represent water using Lewis structures we will get the following:



You will notice that there are two electron pairs that do not take part in bonding. This side of the water molecule has a higher *electron density* than the other side where the hydrogen atoms are bonded. This side of the water molecule is *more negative* than the side where the hydrogen atoms are bonded. We say this side is the delta negative (δ^-) side and the hydrogen side is the delta positive (δ^+) side. This means that one part of the molecule has a slightly *positive* charge (positive pole) and the other part has a slightly *negative* charge (negative pole). We say such a molecule is a **dipole**. It has two poles. Figure 18.2 shows this.

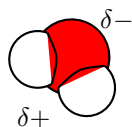


Figure 18.2: Water is a polar molecule

▶ See video: VPbmr at www.everythingscience.co.za

Dissociation of sodium chloride in water ESAFO

It is the polar nature of water that allows ionic compounds to dissolve in it. In the case of sodium chloride (NaCl) for example, the positive sodium ions (Na^+) are attracted to the negative pole of the water molecule, while the negative chloride ions (Cl^-) are attracted to the positive pole of the water molecule. When sodium chloride is dissolved in water, the polar water molecules are able to work their way in between the individual ions in the lattice. The water molecules surround the negative chloride ions and positive sodium ions and pull them away into the solution. This process is called **dissociation**. Note that the positive side of the water molecule will be attracted to the negative chlorine ion and the negative side of the water molecule to the positive sodium ions. A simplified representation of this is shown in Figure 18.3. We say that dissolution of a substance has occurred when a substance dissociates or dissolves. Dissolving is a physical change that takes place. It can be reversed by removing (evaporating) the water.

DEFINITION: Dissociation

Dissociation is a general process in which ionic compounds separate into smaller ions, usually in a reversible manner.

DEFINITION: Dissolution

Dissolution or dissolving is the process where ionic crystals break up into ions in water.

DEFINITION: Hydration

Hydration is the process where ions become surrounded with water molecules.

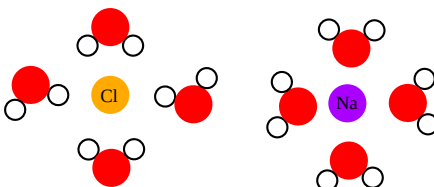
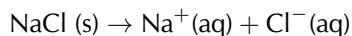
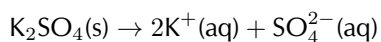


Figure 18.3: Sodium chloride dissolves in water

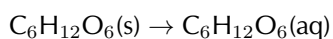
The dissolution of sodium chloride can be represented by the following equation:



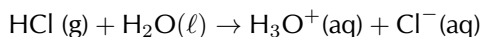
The dissolution of potassium sulphate into potassium and sulphate ions is shown below as another example:



Remember that **molecular** substances (e.g. covalent compounds) may also dissolve, but most will not form ions. One example is glucose.



There are exceptions to this and some molecular substances *will* form ions when they dissolve. Hydrogen chloride for example can ionise to form hydrogen and chloride ions.



You can try dissolving ionic compounds such as potassium permanganate, sodium hydroxide and potassium nitrate in water and observing what happens.

Exercise 18 - 1

1. For each of the following, say whether the substance is ionic or molecular.
 - a. potassium nitrate (KNO_3)
 - b. ethanol ($\text{C}_2\text{H}_5\text{OH}$)
 - c. sucrose (a type of sugar) ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)
 - d. sodium bromide (NaBr)
2. Write a balanced equation to show how each of the following ionic compounds dissociate in water.
 - a. sodium sulphate (Na_2SO_4)
 - b. potassium bromide (KBr)
 - c. potassium permanganate (KMnO_4)
 - d. sodium phosphate (Na_3PO_4)
3. Draw a diagram to show how KCl dissolves in water.

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 0075 (2.) 0076 (3.) 0077

Electrolytes, ionisation and conductivity

 ESAFP

You have learnt that water is a polar molecule and that it can dissolve ionic substances in water. When ions are present in water, the water is able to conduct electricity. The solution is known as an electrolyte.

DEFINITION: *Electrolyte*

An electrolyte is a substance that contains free ions and behaves as an electrically conductive medium.

Because electrolytes generally consist of ions in solution, they are also known as ionic solutions. A strong electrolyte is one where many ions are present in the solution and a weak electrolyte is one where few ions are present. Strong electrolytes are good conductors of electricity and weak electrolytes are weak conductors of electricity. Non-electrolytes do not conduct electricity at all. **Conductivity** in aqueous solutions, is a measure of the ability of water to conduct an electric current. The more **ions** there are in the solution, the higher its conductivity. Also the more ions there are in solution, the stronger the electrolyte.

Factors that affect the conductivity of electrolytes



The conductivity of an electrolyte is therefore affected by the following factors:

- The **concentration of ions** in solution. The higher the concentration of ions in solution, the higher its conductivity will be.
- The **type of substance** that dissolves in water. Whether a material is a strong electrolyte (e.g. potassium nitrate, KNO_3), a weak electrolyte (e.g. acetic acid, CH_3COOH) or a non-electrolyte (e.g. sugar, alcohol, oil) will affect the conductivity of water because the concentration of ions in solution will be different in each case. Strong electrolytes form ions easily, weak electrolytes do not form ions easily and non-electrolytes do not form ions in solution. ▶ See video: VPbon at www.everythingscience.co.za
- **Temperature.** The warmer the solution, the higher the solubility of the material being dissolved and therefore the higher the conductivity as well.

General experiment: *Electrical conductivity*

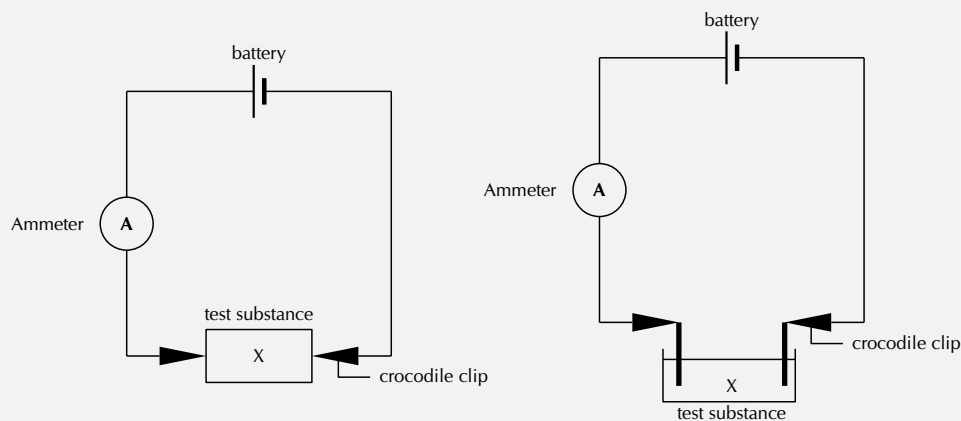
Aim: To investigate the electrical conductivities of different substances and solutions.

Apparatus:

- Solid salt (NaCl) crystals
- different liquids such as distilled water, tap water, seawater, sugar, oil and alcohol
- solutions of salts e.g. NaCl, KBr, CaCl_2 , NH_4Cl
- a solution of an acid (e.g. HCl) and a solution of a base (e.g. NaOH)
- torch cells
- ammeter
- conducting wire, crocodile clips and 2 carbon rods.

Method:

1. Set up the experiment by connecting the circuit as shown in the diagram below. In the diagram, X represents the substance or solution that you will be testing.
2. When you are using the solid crystals, the crocodile clips can be attached directly to each end of the crystal. When you are using solutions, two carbon rods are placed into the liquid and the clips are attached to each of the rods.
3. In each case, complete the circuit and allow the current to flow for about 30 seconds.
4. Observe whether the ammeter shows a reading.



Results: Record your observations in a table similar to the one below:

Test substance	Ammeter reading

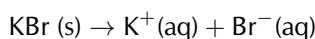
What do you notice? Can you explain these observations?

Conclusions: Solutions that contain free-moving ions are able to conduct electricity because of the movement of charged particles. Solutions that do not contain free-moving ions do not conduct electricity.

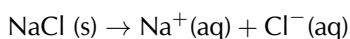
Remember that for electricity to flow, there needs to be a movement of charged particles e.g. ions. With the solid NaCl crystals, there was no flow of electricity recorded on the ammeter. Although the solid is made up of ions, they are held together very tightly within the crystal lattice and therefore no current will flow. Distilled water, oil and alcohol also don't conduct a current because they are *covalent compounds* and therefore do not contain ions.

The ammeter should have recorded a current when the salt solutions and the acid and base solutions were connected in the circuit. In solution, salts *dissociate* into their ions, so that these are free to move in the solution. Look at the following examples:

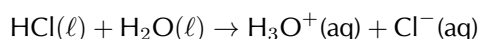
Dissociation of potassium bromide:



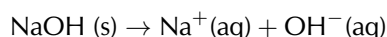
Dissociation of table salt:



Ionisation of hydrochloric acid:



Dissociation of sodium hydroxide:



Precipitation reactions



Sometimes, ions in solution may react with each other to form a new substance that is *insoluble*. This is called a **precipitate**. The reaction is called a precipitation reaction.

► See video: VPbpr at www.everythingscience.co.za

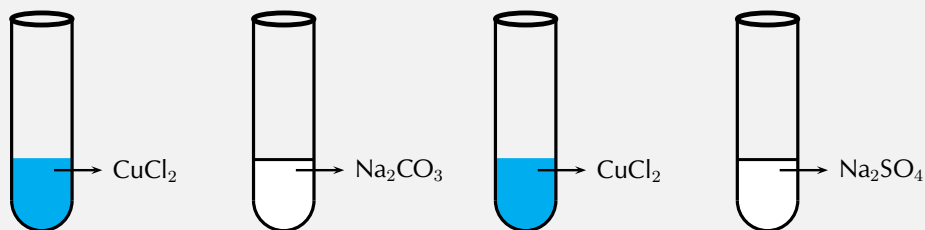
DEFINITION: *Precipitate*

A precipitate is the solid that forms in a solution during a chemical reaction.

General experiment: *The reaction of ions in solution*

Aim: To investigate the reactions of ions in solutions.

Apparatus: 4 test tubes; copper(II) chloride solution; sodium carbonate solution; sodium sulphate solution

**Method:**

1. Prepare 2 test tubes with approximately 5 ml of dilute copper(II) chloride solution in each
2. Prepare 1 test tube with 5 ml sodium carbonate solution
3. Prepare 1 test tube with 5 ml sodium sulphate solution
4. Carefully pour the sodium carbonate solution into one of the test tubes containing copper(II) chloride and observe what happens
5. Carefully pour the sodium sulphate solution into the second test tube containing copper(II) chloride and observe what happens

Results:

1. A light blue precipitate forms when sodium carbonate reacts with copper(II) chloride.
2. No precipitate forms when sodium sulphate reacts with copper(II) chloride. The solution is light blue.

It is important to understand what happened in the previous demonstration. We will look at what happens in each reaction, step by step.

For **reaction 1** you have the following ions in your solution: Cu^{2+} , Cl^- , Na^+ and CO_3^{2-} . A precipitate will form if any combination of cations and anions can become a solid. The following table summarises which combination will form solids (precipitates) in solution.

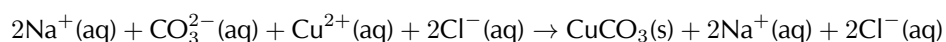
Salt	Solubility
Nitrates	All are soluble
Potassium, sodium and ammonium salts	All are soluble
Chlorides, bromides and iodides	All are soluble except silver, lead(II) and mercury(II) salts (e.g. silver chloride)
Sulphates	All are soluble except lead(II) sulphate, barium sulphate and calcium sulphate
Carbonates	All are insoluble except those of potassium, sodium and ammonium
Compounds with fluorine	Almost all are soluble except those of magnesium, calcium, strontium (II), barium (II) and lead (II)
Perchlorates and acetates	All are soluble
Chlorates	All are soluble except potassium chlorate
Metal hydroxides and oxides	Most are insoluble

Tip

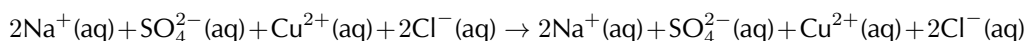
Salts of carbonates, phosphates, oxalates, chromates and sulphides are generally insoluble.

Table 18.1: General rules for the solubility of salts

If you look under carbonates in the table it states that all carbonates are **insoluble** except potassium sodium and ammonium. This means that Na_2CO_3 will dissolve in water or remain in solution, but CuCO_3 will form a precipitate. The precipitate that was observed in the reaction must therefore be CuCO_3 . The balanced chemical equation is:



Note that sodium chloride does not precipitate and we write it as ions in the equation. For **reaction 2** we have Cu^{2+} , Cl^- , Na^+ and SO_4^{2-} in solution. Most chlorides and sulphates are soluble according to the table. The balanced chemical equation is:



Both of these reactions are ion exchange reactions.

Tests for anions

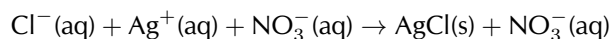


We often want to know which ions are present in solution. If we know which salts precipitate, we can derive tests to identify ions in solution. Given below are a few such tests.

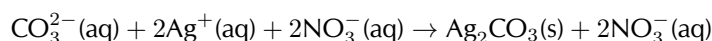
▶ See video: VPbqd at www.everythingscience.co.za

Test for a chloride

Prepare a solution of the unknown salt using distilled water and add a small amount of **silver nitrate** solution. If a white precipitate forms, the salt is either a chloride or a carbonate.

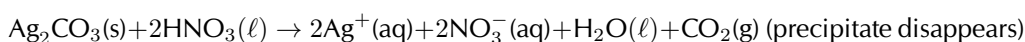


(AgCl is white precipitate)



(Ag₂CO₃ is white precipitate)

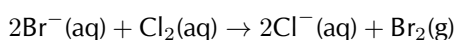
The next step is to treat the precipitate with a small amount of **concentrated nitric acid**. If the precipitate remains unchanged, then the salt is a chloride. If carbon dioxide is formed and the precipitate disappears, the salt is a carbonate.



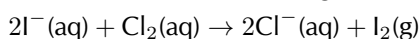
Test for bromides and iodides

As was the case with the chlorides, the bromides and iodides also form precipitates when they are reacted with silver nitrate. Silver chloride is a white precipitate, but the silver bromide and silver iodide precipitates are both pale yellow. To determine whether the precipitate is a bromide or an iodide, we use chlorine water and carbon tetrachloride (CCl₄).

Chlorine water frees bromine gas from the bromide and colours the carbon tetrachloride a reddish brown.

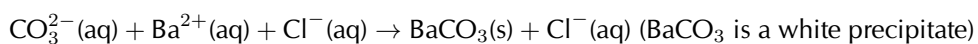
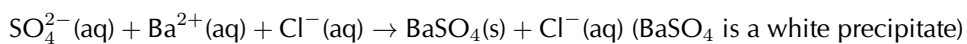


Chlorine water frees iodine gas from an iodide and colours the carbon tetrachloride purple.

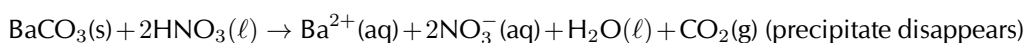


Test for a sulphate

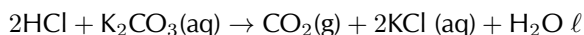
Add a small amount of barium chloride solution to a solution of the test salt. If a white precipitate forms, the salt is either a sulphate or a carbonate.



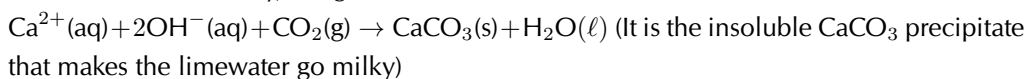
If the precipitate is treated with nitric acid, it is possible to distinguish whether the salt is a sulphate or a carbonate (as in the test for a chloride).

**Test for a carbonate**

If a sample of the dry salt is treated with a small amount of acid, the production of carbon dioxide is a positive test for a carbonate.



If the gas is passed through limewater (an aqueous solution of calcium hydroxide) and the solution becomes milky, the gas is carbon dioxide.

**Exercise 18 - 2**

1. Silver nitrate (AgNO_3) reacts with potassium chloride (KCl) and a white precipitate is formed.
 - a. Write a balanced equation for the reaction that takes place. Include the state symbols.
 - b. What is the name of the insoluble salt that forms?
 - c. Which of the salts in this reaction are soluble?
2. Barium chloride reacts with sulphuric acid to produce barium sulphate and hydrochloric acid.
 - a. Write a balanced equation for the reaction that takes place. Include the state symbols.
 - b. Does a precipitate form during the reaction?
 - c. Describe a test that could be used to test for the presence of barium sulphate in the products.

3. A test tube contains a clear, colourless salt solution. A few drops of silver nitrate solution are added to the solution and a pale yellow precipitate forms. Chlorine water and carbon tetrachloride were added, which resulted in a purple solution. Which one of the following salts was dissolved in the original solution? Write the balanced equation for the reaction that took place between the salt and silver nitrate.

- a. NaI
- b. KCl
- c. K_2CO_3
- d. Na_2SO_4

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 0078 (2.) 0079 (3.) 007a

Other types of reactions

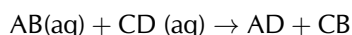
 ESAFT

We will look at two types of reactions that occur in aqueous solutions. These are ion-exchange reactions and redox reactions. Ion exchange reactions include precipitation reactions, gas forming reactions and acid-base reactions. Redox reactions are electron transfer reactions. It is important to remember the difference between these two types of reactions. In ion exchange reactions ions are exchanged, in electron transfer reactions electrons are transferred. These terms will be explained further in the following sections.

Ion exchange reactions

 ESAFU

Ion exchange reactions can be represented by:



Either AD or CB may be a solid or a gas. When a solid forms this is known as a precipitation reaction. If a gas is formed then this may be called a gas forming reaction. Acid-base reactions are a special class of ion exchange reactions and we will look at them separately.

The formation of a precipitate or a gas helps to make the reaction happen. We say that the

reaction is driven by the formation of a precipitate or a gas. All chemical reactions will only take place if there is something to make them happen. For some reactions this happens easily and for others it is harder to make the reaction occur.

FACT

Ion exchange reactions are used in ion exchange chromatography.

Ion exchange chromatography is used to purify water and as a means of softening water. Often when chemists talk about ion exchange, they mean ion exchange chromatography.

DEFINITION: *Ion exchange reaction*

A type of reaction where the positive ions exchange their respective negative ions due to a driving force.

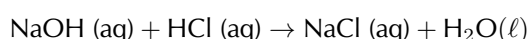
We have already looked at precipitation reactions.

Gas forming reactions

These reactions are similar to precipitation reactions with the exception that instead of a precipitate forming, a gas is formed instead. An example of a gas forming reaction is sodium carbonate in hydrochloric acid. The balanced equation for this reaction is:

**Acid-base reactions**

Acid-base reactions take place between acids and bases. In general, the products will be water and a salt (i.e. an ionic compound). An example of this type of reaction is:



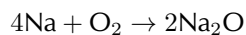
This is an special case of an ion exchange reaction since the sodium in the sodium hydroxide swaps places with the hydrogen in the hydrogen chloride forming sodium chloride. At the same time the hydroxide and the hydrogen combine to form water.

Redox reactions



Redox reactions involve the exchange of electrons. One ion loses electrons and becomes more positive, while the other ion gains electrons and becomes more negative. To decide if a redox reaction has occurred we look at the charge of the atoms, ions or molecules involved. If one of them has become more positive and the other one has become more

negative then a redox reaction has occurred. For example, sodium metal is oxidised to form sodium oxide (and sometimes sodium peroxide as well). The balanced equation for this is:



In the above reaction sodium and oxygen are both neutral and so have no charge. In the products however, the sodium atom has a charge of +1 and the oxygen atom has a charge of -2. This tells us that the sodium has lost electrons and the oxygen has gained electrons. Since one species has become more positive and one more negative we can conclude that a redox reaction has occurred. We could also say that electrons have been transferred from one species to the other. (In this case the electrons were transferred from the sodium to the oxygen).

General experiment: Demonstration: Oxidation of sodium metal

You will need a Bunsen burner, a small piece of sodium metal and a metal spatula. Light the Bunsen burner. Place the sodium metal on the spatula. Place the sodium in the flame. When the reaction finishes, you should observe a white powder on the spatula. This is a mixture of sodium oxide (Na_2O) and sodium peroxide (Na_2O_2).

Warning:

Sodium metal is very reactive. Sodium metal reacts vigorously with water and should never be placed in water. Be very careful when handling sodium metal.



General experiment: Reaction types

Aim: To use experiments to determine what type of reaction occurs.

Apparatus: Soluble salts (e.g. potassium nitrate, ammonium chloride, sodium carbonate, silver nitrate, sodium bromide); hydrochloric acid (HCl); sodium hydroxide (NaOH); bromothymol blue; zinc metal; copper (II) sulphate; beakers; test-tubes

Method:

1. For each of the salts, dissolve a small amount in water and observe what happens.
2. Try dissolving pairs of salts (e.g. potassium nitrate and sodium carbonate) in water and observe what happens.
3. Dissolve some sodium carbonate in hydrochloric acid and observe what happens.
4. Carefully measure out 20cm^3 of sodium hydroxide into a beaker.
5. Add some bromothymol blue to the sodium hydroxide
6. Carefully add a few drops of hydrochloric acid to the sodium hydroxide and swirl. Repeat until you notice the colour change.
7. Place the zinc metal into the copper sulphate solution and observe what happens.

Results: Answer the following questions:

1. What did you observe when you dissolved each of the salts in water?
2. What did you observe when you dissolved pairs of salts in the water?
3. What did you observe when you dissolved sodium carbonate in hydrochloric acid?
4. Why do you think we used bromothymol blue when mixing the hydrochloric acid and the sodium hydroxide? Think about the kind of reaction that occurred.
5. What did you observe when you placed the zinc metal into the copper sulphate?
6. Classify each reaction as either precipitation, gas forming, acid-base or redox.
7. What makes each reaction happen (i.e. what is the driving force)? Is it the formation of a precipitate or something else?
8. What criteria would you use to determine what kind of reaction occurs?
9. Try to write balanced chemical equations for each reaction

Conclusion: We can see how we can classify reactions by performing experiments.

In the experiment above, you should have seen how each reaction type differs from the others. For example, a gas forming reaction leads to bubbles in the solution, a precipitation reaction leads to a precipitate forming, an acid-base reaction can be seen by adding a suitable indicator and a redox reaction can be seen by one metal disappearing and a deposit forming in the solution.

Chapter 18 | Summary

See the summary presentation (▶ Presentation: VPenn at www.everythingscience.co.za)

- The **polar** nature of water means that **ionic compounds** dissociate easily in aqueous solution into their component ions.
- Dissociation is a general process in which ionic compounds separate into smaller ions, usually in a reversible manner.
- Dissolution or dissolving is the process where ionic crystals break up into ions in water.
- Hydration is the process where ions become surrounded with water molecules.
- **Conductivity** is a measure of a solution's ability to conduct an electric current.
- An **electrolyte** is a substance that contains free ions and is therefore able to conduct an electric current. Electrolytes can be divided into **strong** and **weak** electrolytes, based on the extent to which the substance ionises in solution.
- A **non-electrolyte** cannot conduct an electric current because it does not contain free ions.
- The **type of substance**, the **concentration of ions** and the **temperature** of the solution affect its conductivity.
- There are three main types of reactions that occur in aqueous solutions. These are precipitation reactions, acid-base reactions and redox reactions.
- Precipitation and acid-base reactions are sometimes known as ion exchange reactions. Ion exchange reactions also include gas forming reactions. Ion exchange reactions are a type of reaction where the positive ions exchange their respective negative ions due to a driving force.
- A **precipitate** is formed when ions in solution react with each other to form an insoluble product. Solubility rules help to identify the precipitate that has been formed.
- A number of tests can be used to identify whether certain **anions** (chlorides, bromides, iodides, carbonates, sulphates) are present in a solution.
- An acid-base reaction is one in which an acid reacts with a base to form a salt and water.

- A redox reaction is one in which electrons are transferred from one substance to another.

Chapter 18**End of chapter exercises**

1. Give one word for each of the following descriptions:
 - a. the change in phase of water from a gas to a liquid
 - b. a charged atom
 - c. a term used to describe the mineral content of water
 - d. a gas that forms sulphuric acid when it reacts with water
2. Match the information in column A with the information in column B by writing only the letter (A to I) next to the question number (1 to 7)

Column A	Column B
1. A polar molecule	A. H_2SO_4
2. Molecular solution	B. CaCO_3
3. Mineral that increases water hardness	C. NaOH
4. Substance that increases the hydrogen ion concentration	D. salt water
5. A strong electrolyte	E. calcium
6. A white precipitate	F. carbon dioxide
7. A non-conductor of electricity	G. potassium nitrate
	H. sugar water
	I. O_2

3. Explain the difference between a weak electrolyte and a strong electrolyte. Give a generalised equation for each.
4. What factors affect the conductivity of water? How do each of these affect the conductivity?
5. For each of the following substances state whether they are molecular or ionic. If they are ionic, give a balanced reaction for the dissociation in water.
 - a. methane (CH_4)
 - b. potassium bromide

- c. carbon dioxide
 - d. hexane (C_6H_{14})
 - e. lithium fluoride (LiF)
 - f. magnesium chloride
6. Three test tubes (X, Y and Z) each contain a solution of an unknown potassium salt. The following observations were made during a practical investigation to identify the solutions in the test tubes:
- A: A white precipitate formed when silver nitrate (AgNO_3) was added to test tube Z.
- B: A white precipitate formed in test tubes X and Y when barium chloride (BaCl_2) was added.
- C: The precipitate in test tube X dissolved in hydrochloric acid (HCl) and a gas was released.
- D: The precipitate in test tube Y was insoluble in hydrochloric acid.
- a. Use the above information to identify the solutions in each of the test tubes X, Y and Z.
 - b. Write a chemical equation for the reaction that took place in test tube X before hydrochloric acid was added.

(DoE Exemplar Paper 2 2007)

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 007b (2.) 007c (3.) 007d (4.) 007e (5.) 007f (6.) 007g

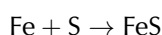
Quantitative aspects of chemical change

19

Atomic mass and the mole

 ESAFW

An equation for a chemical reaction can provide us with a lot of useful information. It tells us what the reactants and the products are in the reaction, and it also tells us the ratio in which the reactants combine to form products. Look at the equation below:



In this reaction, every atom of iron (Fe) will react with a single atom of sulphur (S) to form iron sulphide (FeS). However, what the equation does not tell us, is the **quantities** or the **amount** of each substance that is involved. You may for example be given a small sample of iron for the reaction. How will you know how many atoms of iron are in this sample? And how many atoms of sulphur will you need for the reaction to use up all the iron you have? Is there a way of knowing what mass of iron sulphide will be produced at the end of the reaction? These are all very important questions, especially when the reaction is an industrial one, where it is important to know the quantities of reactants that are needed, and the quantity of product that will be formed. This chapter will look at how to quantify the changes that take place in chemical reactions.

See introductory video: (📺 Video: VPbrj at www.everythingscience.co.za)

The Mole

 ESAFX

Sometimes it is important to know exactly how many particles (e.g. atoms or molecules) are in a sample of a substance, or what quantity of a substance is needed for a chemical reaction to take place.

The amount of substance is so important in chemistry that it is given its own name, which is the mole.

DEFINITION: Mole

The mole (abbreviation “mol”) is the SI (Standard International) unit for “amount of substance”.

The mole is a counting unit just like hours or days. We can easily count one second or one minute or one hour. If we want bigger units of time, we refer to days, months and years. Even longer time periods are centuries and millennia. The mole is even bigger than these numbers. The mole is 602 204 500 000 000 000 000 000 or $6,022 \times 10^{23}$ particles. This is a very big number! We call this number Avogadro’s number.

DEFINITION: Avogadro’s number

The number of particles in a mole, equal to $6,022 \times 10^{23}$.

If we had this number of cold drink cans, then we could cover the surface of the earth to a depth of over 300 km! If you could count atoms at a rate of 10 million per second, then it would take you 2 billion years to count the atoms in one mole!

We use Avogadro’s number and the mole in chemistry to help us quantify what happens in chemical reaction. The mole is a very special number. If we measure 12,0 g of carbon we have one mole or $6,022 \times 10^{23}$ carbon atoms. 63,5 g of copper is one mole of copper or $6,022 \times 10^{23}$ copper atoms. In fact, if we measure the relative atomic mass of any element on the periodic table, we have one mole of that element.

▶ See video: VPbsf at www.everythingscience.co.za

Exercise 19 - 1

1. How many atoms are there in:

- | | |
|--------------------------|------------------------|
| a. 1 mole of a substance | d. 24,3 g of magnesium |
| b. 2 moles of calcium | e. 24,0 g of carbon |
| c. 5 moles of phosphorus | |

2. Complete the following table:

FACT

The original hypothesis that was proposed by Amadeo Avogadro was that “equal volumes of gases, at the same temperature and pressure, contain the same number of molecules”. His ideas were not accepted by the scientific community and it was only four years after his death, that his original hypothesis was accepted and that it became known as “Avogadro’s Law”. In honour of his contribution to science, the number of particles in one mole was named *Avogadro’s number*.

Element	Relative atomic mass (u)	Sample mass (g)	Number of moles in the sample
Hydrogen	1,01	1,01	
Magnesium	24,3	24,3	
Carbon	12,0	24,0	
Chlorine	35,45	70,9	
Nitrogen	14,0	42,0	

 More practice
  video solutions
  or help at www.everythingscience.co.za

(1.) 007h (2.) 007i

Molar Mass



DEFINITION: Molar mass

Molar mass (M) is the mass of 1 mole of a chemical substance. The unit for molar mass is **grams per mole** or $\text{g} \cdot \text{mol}^{-1}$.

Note

You may sometimes see the molar mass written as M_m . We will use M in this book, but you should be aware of the alternate notation.

You will remember that when the mass, in grams, of an element is equal to its relative atomic mass, the sample contains one mole of that element. This mass is called the **molar mass** of that element.

It is worth remembering the following: On the periodic table, the relative atomic mass that is shown can be interpreted in two ways.

1. The mass (in grams) of a *single, average atom* of that element relative to the mass of an atom of carbon.
2. The average atomic mass of all the isotopes of that element. This use is the *relative atomic mass*.
3. The mass of *one mole of the element*. This third use is the molar mass of the element.

Element	Relative atomic mass (u)	Molar mass ($\text{g} \cdot \text{mol}^{-1}$)	Mass of one mole of the element (g)
Magnesium	24,3	24,3	24,3
Lithium	6,94	6,94	6,94
Oxygen	16,0	16,0	16,0
Nitrogen	14,0	14,0	14,0
Iron	55,8	55,8	55,8

Table 19.1: The relationship between relative atomic mass, molar mass and the mass of one mole for a number of elements.

Example 1: Calculating the number of moles from mass

QUESTION

Calculate the number of moles of iron (Fe) in an 11,7 g sample.

SOLUTION

Step 1 : Find the molar mass of iron

If we look at the periodic table, we see that the molar mass of iron is $55,8 \text{ g} \cdot \text{mol}^{-1}$. This means that 1 mole of iron will have a mass of 55,8 g.

Step 2 : Find the mass of iron

If 1 mole of iron has a mass of 55,8 g, then: the number of moles of iron in 11,7 g must be:

$$\begin{aligned}n &= \frac{11,7 \text{ g}}{55,8 \text{ g} \cdot \text{mol}^{-1}} \\&= \frac{11,7 \text{ g} \cdot \text{mol}}{55,8 \text{ g}} \\&= 2 \text{ mol}\end{aligned}$$

There are 2 moles of iron in the sample.

Example 2: Calculating mass from moles**QUESTION**

You have a sample that contains 5 moles of zinc.

- a. What is the mass of the zinc in the sample?
- b. How many atoms of zinc are in the sample?

SOLUTION**Step 1 : Find the molar mass of zinc**

Molar mass of zinc is $65,4 \text{ g} \cdot \text{mol}^{-1}$, meaning that 1 mole of zinc has a mass of 65,4 g.

Step 2 : Find the mass

If 1 mole of zinc has a mass of 65,4 g, then 5 moles of zinc has a mass of: $65,4 \text{ g} \times 5 \text{ mol} = 327 \text{ g}$ (answer to a)

Step 3 : Find the number of atoms

$5 \text{ mol} \times 6,022 \times 10^{23} \text{ atoms} \cdot \text{mol}^{-1} = 3,011 \times 10^{23} \text{ atoms}$ (answer to b)

Exercise 19 - 2

1. Give the molar mass of each of the following elements:
 - a. hydrogen gas
 - b. nitrogen gas
 - c. bromine gas
2. Calculate the number of moles in each of the following samples:
 - a. 21,6 g of boron (B)
 - b. 54,9 g of manganese (Mn)
 - c. 100,3 g of mercury (Hg)
 - d. 50 g of barium (Ba)
 - e. 40 g of lead (Pb)



More practice



video solutions

or help at www.everythingscience.co.za

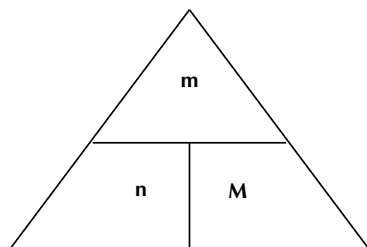
(1.) 007j (2.) 007k

An equation to calculate moles and mass ESAFZ

We can calculate molar mass as follows: $\text{molar mass (M)} = \frac{\text{mass (g)}}{\text{mole (mol)}}$
This can be rearranged to give the number of moles:

$$n = \frac{m}{M}$$

The following diagram may help to remember the relationship between these three variables. You need to imagine that the horizontal line is like a division sign and that the vertical line is like a multiplication sign. So, for example, if you want to calculate M , then the remaining two letters in the triangle are m and n and m is above n with a division sign between them. Your calculation will then be $M = \frac{m}{n}$

**Tip**

Remember that when you use the equation $n = \frac{m}{M}$, the mass is always in grams (g) and molar mass is in grams per mol ($\text{g} \cdot \text{mol}^{-1}$). Always write the units next to any number you use in a formula or sum.

Example 3: Calculating moles from mass**QUESTION**

Calculate the number of moles of copper there are in a sample that with a mass of 127 g.

SOLUTION

Step 1 : **Write down the equation**

$$n = \frac{m}{M}$$

Step 2 : **Find the moles**

$$n = \frac{127 \text{ g}}{63,5 \text{ g} \cdot \text{mol}^{-1}} = 2 \text{ mol}$$

There are 2 moles of copper in the sample.

Example 4: Calculating atoms from mass

QUESTION

Calculate the number of atoms there are in a sample of aluminium that with a mass of 81 g.

SOLUTION

Step 1 : **Find the number of moles**

$$n = \frac{m}{M} = \frac{81 \text{ g}}{27,0 \text{ g} \cdot \text{mol}^{-1}} = 3 \text{ mol}$$

Step 2 : **Find the number of atoms**

Number of atoms in 3 mol aluminium = $3 \times 6,022 \times 10^{23}$

There are $1,8069 \times 10^{24}$ aluminium atoms in a sample of 81 g.

Exercise 19 - 3

1. Calculate the number of moles in each of the following samples:

- a. 5,6 g of calcium
- b. 0,02 g of manganese
- c. 40 g of aluminium

2. A lead sinker has a mass of 5 g.

- a. Calculate the number of moles of lead the sinker contains.
- b. How many lead atoms are in the sinker?

3. Calculate the mass of each of the following samples:

- a. 2,5 mol magnesium
- b. 12 mol lithium
- c. $4,5 \times 10^{25}$ atoms of silicon

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 007m (2.) 007n (3.) 007p

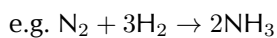
Compounds



So far, we have only discussed moles, mass and molar mass in relation to *elements*. But what happens if we are dealing with a compound? Do the same concepts and rules apply? The answer is yes. However, you need to remember that all your calculations will apply to the *whole compound*. So, when you calculate the **molar mass** of a covalent compound, you will need to add the molar mass of each atom in that compound. The number of moles will also apply to the whole molecule. For example, if you have one mole of nitric acid (HNO_3) the molar mass is $63,01 \text{ g} \cdot \text{mol}^{-1}$ and there are $6,022 \times 10^{23}$ molecules of nitric acid. For network structures we have to use the formula mass. This is the mass of all the atoms in **one formula unit** of the compound. For example, one mole of sodium chloride (NaCl) has a formula mass of $63,01 \text{ g} \cdot \text{mol}^{-1}$ and there are $6,022 \times 10^{23}$ molecules of sodium chloride in one formula unit.

In a balanced chemical equation, the number that is written in front of the element or compound, shows the **mole ratio** in which the reactants combine to form a product. If there are no numbers in front of the element symbol, this means the number is '1'.

▶ See video: VPezc at www.everythingscience.co.za



In this reaction, 1 mole of nitrogen molecules reacts with 3 moles of hydrogen molecules to produce 2 moles of ammonia molecules.

Example 5: Calculating molar mass

QUESTION

Calculate the molar mass of H_2SO_4 .

SOLUTION

Step 1 : **Give the molar mass for each element**

Hydrogen = $1,01 \text{ g} \cdot \text{mol}^{-1}$

Sulphur = $32,1 \text{ g} \cdot \text{mol}^{-1}$

Oxygen = $16,0 \text{ g} \cdot \text{mol}^{-1}$

Step 2 : **Work out the molar mass of the compound**

$$M_{(\text{H}_2\text{SO}_4)} = (2 \times 1,01 \text{ g} \cdot \text{mol}^{-1}) + (32,1 \text{ g} \cdot \text{mol}^{-1}) + (4 \times 16,0 \text{ g} \cdot \text{mol}^{-1}) = 98,12 \text{ g} \cdot \text{mol}^{-1}$$

Example 6: Calculating moles from mass

QUESTION

Calculate the number of moles in 1 kg of MgCl_2 .

SOLUTION

Step 1 : **Convert mass into grams**

$$m = 1 \text{ kg} \times 1\,000 = 1\,000 \text{ g}$$

Step 2 : **Calculate the molar mass**

$$M_{(\text{MgCl}_2)} = 24,3 \text{ g} \cdot \text{mol}^{-1} + (2 \times 35,45 \text{ g} \cdot \text{mol}^{-1}) = 95,2 \text{ g} \cdot \text{mol}^{-1}$$

Step 3 : **Find the number of moles**

$$n = \frac{1\,000 \text{ g}}{95,2 \text{ g} \cdot \text{mol}^{-1}} = 10,5 \text{ mol}$$

There are 10,5 moles of magnesium chloride in a 1 kg sample.

Group Discussion: Understanding moles, molecules and Avogadro's number

Divide into groups of three and spend about 20 minutes answering the following questions together:

1. What are the units of the mole? Hint: Check the definition of the mole.
 2. You have a 46 g sample of nitrogen dioxide (NO_2)
 - a. How many **moles** of NO_2 are there in the sample?
 - b. How many moles of nitrogen atoms are there in the sample?
 - c. How many moles of oxygen atoms are there in the sample?
 - d. How many **molecules** of NO_2 are there in the sample?
 - e. What is the difference between a mole and a molecule?
 3. The exact size of **Avogadro's number** is sometimes difficult to imagine.
 - a. Write down Avogadro's number without using scientific notation.
 - b. How long would it take to count to Avogadro's number? You can assume that you can count two numbers in each second.
-

Exercise 19 - 4

1. Calculate the molar mass of the following chemical compounds:

- a. KOH
 - b. FeCl_3
 - c. $\text{Mg}(\text{OH})_2$
2. How many moles are present in:
- a. 10 g of Na_2SO_4
 - b. 34 g of $\text{Ca}(\text{OH})_2$
 - c. $2,45 \times 10^{23}$ molecules of CH_4 ?
3. For a sample of 0,2 moles of magnesium bromide (MgBr_2), calculate:
- a. the number of moles of Mg^{2+} ions
 - b. the number of moles of Br^- ions
4. You have a sample containing 3 mol of calcium chloride.
- a. What is the chemical formula of calcium chloride?
 - b. How many calcium atoms are in the sample?
5. Calculate the mass of:
- a. 3 moles of NH_4OH
 - b. 4,2 moles of $\text{Ca}(\text{NO}_3)_2$

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 007q (2.) 007r (3.) 007s (4.) 007t (5.) 007u

Composition



Knowing either the empirical or molecular formula of a compound, can help to determine its composition in more detail. The opposite is also true. Knowing the *composition* of a substance can help you to determine its formula. There are four different types of composition problems that you might come across:

1. Problems where you will be **given the formula** of the substance and asked to **calculate the percentage by mass** of each element in the substance.
2. Problems where you will be **given the percentage composition** and asked to **calculate the formula**.
3. Problems where you will be **given the products** of a chemical reaction and asked to **calculate the formula** of one of the reactants. These are often referred to as combustion analysis problems.

4. Problems where you will be asked to **find** number of moles of **waters of crystallisation**.

The following worked examples will show you how to do each of these.

📺 See video: VPbue at www.everythingscience.co.za

Example 7: Calculating the percentage by mass of elements in a compound

QUESTION

Calculate the percentage that each element contributes to the overall mass of sulphuric acid (H_2SO_4).

SOLUTION

Step 1 : **Calculate the molar masses**

$$\text{Hydrogen} = 2 \times 1,01 = 2,02 \text{ g} \cdot \text{mol}^{-1}$$

$$\text{Sulphur} = 32,1 \text{ g} \cdot \text{mol}^{-1}$$

$$\text{Oxygen} = 4 \times 16,0 = 64,0 \text{ g} \cdot \text{mol}^{-1}$$

Step 2 : **Use the calculations in the previous step to calculate the molecular mass of sulphuric acid.**

$$\text{Mass} = 2,02 \text{ g} \cdot \text{mol}^{-1} + 32,1 \text{ g} \cdot \text{mol}^{-1} + 64,0 \text{ g} \cdot \text{mol}^{-1} = 98,12 \text{ g} \cdot \text{mol}^{-1}$$

Step 3 : **Use the equation**

$$\text{Percentage by mass} = \frac{\text{atomic mass}}{\text{molecular mass of } \text{H}_2\text{SO}_4} \times 100\%$$

Hydrogen

$$\frac{2,02 \text{ g} \cdot \text{mol}^{-1}}{98,12 \text{ g} \cdot \text{mol}^{-1}} \times 100\% = 2,0587\%$$

Sulphur

$$\frac{32,1 \text{ g} \cdot \text{mol}^{-1}}{98,12 \text{ g} \cdot \text{mol}^{-1}} \times 100\% = 32,7150\%$$

Oxygen

$$\frac{64,0 \text{ g} \cdot \text{mol}^{-1}}{98,12 \text{ g} \cdot \text{mol}^{-1}} \times 100\% = 65,2263\%$$

(You should check at the end that these percentages add up to 100%!)

In other words, in one molecule of sulphuric acid, hydrogen makes up 2,06% of the mass of the compound, sulphur makes up 32,71% and oxygen makes up 65,23%.

Example 8: *Determining the empirical formula of a compound*

QUESTION

A compound contains 52,2% carbon (C), 13,0% hydrogen (H) and 34,8% oxygen (O). Determine its empirical formula.

SOLUTION

Step 1 : **Give the masses**

Carbon = 52,2 g, hydrogen = 13,0 g and oxygen = 34,8 g

Step 2 : **Calculate the number of moles**

$$n = \frac{m}{M}$$

Therefore:

$$n(\text{Carbon}) = \frac{52,2 \text{ g}}{12,0 \text{ g} \cdot \text{mol}^{-1}} = 4,35 \text{ mol}$$

$$n(\text{Hydrogen}) = \frac{13,0 \text{ g}}{1,01 \text{ g} \cdot \text{mol}^{-1}} = 12,871 \text{ mol}$$

$$n(\text{Oxygen}) = \frac{34,8 \text{ g}}{16,0 \text{ g} \cdot \text{mol}^{-1}} = 2,175 \text{ mol}$$

Step 3 : **Find the smallest number of moles**

Use the ratios of molar numbers calculated above to find the empirical formula.

$$\text{units in empirical formula} = \frac{\text{moles of this element}}{\text{smallest number of moles}}$$

In this case, the smallest number of moles is 2,175. Therefore:
Carbon

$$\frac{4,35}{2,175} = 2$$

Hydrogen

$$\frac{12,871}{2,175} = 6$$

Oxygen

$$\frac{2,175}{2,175} = 1$$

Therefore the empirical formula of this substance is: C₂H₆O.

Example 9: Determining the formula of a compound

QUESTION

207 g of lead combines with oxygen to form 239 g of a lead oxide. Use this information to work out the formula of the lead oxide (Relative atomic masses: Pb = 207,2 u and O = 16,0 u).

SOLUTION

Step 1 : **Find the mass of oxygen**

$$239 \text{ g} - 207 \text{ g} = 32 \text{ g}$$

Step 2 : **Find the moles of oxygen**

$$n = \frac{m}{M}$$

Lead

$$n = \frac{207 \text{ g}}{207,2 \text{ g} \cdot \text{mol}^{-1}} = 1 \text{ mol}$$

Oxygen

$$n = \frac{32 \text{ g}}{16,0 \text{ g} \cdot \text{mol}^{-1}} = 2 \text{ mol}$$

Step 3 : **Find the mole ratio**

The mole ratio of Pb : O in the product is 1 : 2, which means that for every atom of lead, there will be two atoms of oxygen. The formula of the compound is PbO₂.

Example 10: Empirical and molecular formula

QUESTION

Vinegar, which is used in our homes, is a dilute form of acetic acid. A sample of acetic acid has the following percentage composition: 39,9% carbon, 6,7% hydrogen and 53,4% oxygen.

1. Determine the empirical formula of acetic acid.
2. Determine the molecular formula of acetic acid if the molar mass of acetic acid is 60,06 g · mol⁻¹.

SOLUTION

Step 1 : **Find the mass**

In 100 g of acetic acid, there is 39,9 g C, 6,7 g H and 53,4 g O

Step 2 : **Find the moles**

$$n = \frac{m}{M}$$

$$n_{\text{C}} = \frac{39,9 \text{ g}}{12,0 \text{ g} \cdot \text{mol}^{-1}} = 3,325 \text{ mol}$$

$$n_{\text{H}} = \frac{6,7 \text{ g}}{1,01 \text{ g} \cdot \text{mol}^{-1}} = 6,6337 \text{ mol}$$

$$n_{\text{O}} = \frac{53,4 \text{ g}}{16,0 \text{ g} \cdot \text{mol}^{-1}} = 3,3375 \text{ mol}$$

Step 3 : **Find the empirical formula**

C : H : O

3,325 : 6,6337 : 3,3375

1 : 2 : 1

Empirical formula is CH_2O

Step 4 : **Find the molecular formula**

The molar mass of acetic acid using the empirical formula is $30,02 \text{ g} \cdot \text{mol}^{-1}$. However the question gives the molar mass as $60,06 \text{ g} \cdot \text{mol}^{-1}$. Therefore the actual number of moles of each element must be double what it is in the empirical formula ($\frac{60,06}{30,02} = 2$). The molecular formula is therefore $\text{C}_2\text{H}_4\text{O}_2$ or CH_3COOH

Example 11: Waters of crystallisation

QUESTION

Aluminium trichloride (AlCl_3) is an ionic substance that forms crystals in the solid phase. Water molecules may be trapped inside the crystal lattice. We represent this as: $\text{AlCl}_3 \cdot n\text{H}_2\text{O}$. Carine heated some aluminium trichloride crystals until all the water had evaporated and found that the mass after heating was 2,8 g. The mass before heating was 5 g. What is the number of moles of water molecules in the

aluminium trichloride before heating?

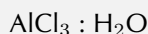
SOLUTION

Step 1 : Find the number of water molecules

We first need to find n , the number of water molecules that are present in the crystal. To do this we first note that the mass of water lost is $5\text{ g} - 2,8\text{ g} = 2,2\text{ g}$.

Step 2 : Find the mass ratio

The mass ratio is:



$$2,8 : 2,2$$

Step 3 : Find the mole ratio

To work out the mole ratio we divide the mass ratio by the molecular mass of each species:

$$\begin{array}{c} \text{AlCl}_3 : \text{H}_2\text{O} \\ \frac{2,8\text{ g}}{133,35\text{ g} \cdot \text{mol}^{-1}} : \frac{2,2\text{ g}}{18,02\text{ g} \cdot \text{mol}^{-1}} \\ 0,02099\dots : 0,12\dots \end{array}$$

Next we convert the ratio to whole numbers by dividing both sides by the smaller amount:

$$\begin{array}{c} \text{AlCl}_3 : \text{H}_2\text{O} \\ 0,020997375 : 0,12208657 \\ \frac{0,021}{0,021} : \frac{0,122}{0,021} \\ 1 : 6 \end{array}$$

The mole ratio of aluminium trichloride to water is: 1 : 6

Step 4 : Write the final answer

And now we know that there are 6 moles of water molecules in the crystal. The formula is $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$.

We can perform experiments to determine the composition of substances. For example, blue copper sulphate (CuSO_4) crystals contain water. On heating the waters of crystallisation evaporate and the blue crystals become white. By weighing the starting and ending products, we can determine the amount of water that is in copper sulphate. Another example is reducing copper oxide to copper.

Exercise 19 - 5

1. Calcium chloride is produced as the product of a chemical reaction.
 - a. What is the formula of calcium chloride?
 - b. What is the percentage mass of each of the elements in a molecule of calcium chloride?
 - c. If the sample contains 5 g of calcium chloride, what is the mass of calcium in the sample?
 - d. How many moles of calcium chloride are in the sample?
2. 13 g of zinc combines with 6,4 g of sulphur.
 - a. What is the empirical formula of zinc sulphide?
 - b. What mass of zinc sulphide will be produced?
 - c. What is the percentage mass of each of the elements in zinc sulphide?
 - d. The molar mass of zinc sulphide is found to be $97,44 \text{ g} \cdot \text{mol}^{-1}$. Determine the molecular formula of zinc sulphide.
3. A calcium mineral consisted of 29,4% calcium, 23,5% sulphur and 47,1% oxygen by mass. Calculate the empirical formula of the mineral.
4. A chlorinated hydrocarbon compound was analysed and found to consist of 24,24% carbon, 4,04% hydrogen and 71,72% chlorine. From another experiment the molecular mass was found to be $99 \text{ g} \cdot \text{mol}^{-1}$. Deduce the empirical and molecular formula.
5. Magnesium sulphate has the formula $\text{MgSO}_4 \cdot n \text{H}_2\text{O}$. A sample containing 5,0 g of magnesium sulphate was heated until all the water had evaporated. The final mass was found to be 2,6 g. How many water molecules were in the original sample?

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 007v (2.) 007w (3.) 007x (4.) 007y (5.) 007z

Amount of substance

www ESAGC

Molar Volumes of Gases

www ESAGD

DEFINITION: *Molar volume of gases*

One mole of gas occupies $22,4 \text{ dm}^3$ at standard temperature and pressure.

Note

Standard temperature and pressure (S.T.P.) is defined as a temperature of $273,15 \text{ K}$ and a pressure of $0,986 \text{ atm}$.

This applies to any gas that is at standard temperature and pressure. In grade 11 you will learn more about this and the gas laws.

For example, 2 mol of H_2 gas will occupy a volume of $44,8 \text{ dm}^3$ at standard temperature and pressure (S.T.P.). and $67,2 \text{ dm}^3$ of ammonia gas (NH_3) contains 3 mol of ammonia.

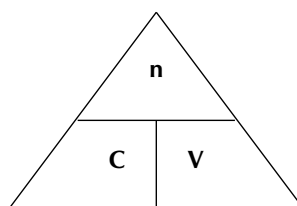
Molar concentrations of liquids

www ESAGE

A typical solution is made by dissolving some solid substance in a liquid. The amount of substance that is dissolved in a given volume of liquid is known as the **concentration** of the liquid. Mathematically, concentration (C) is defined as moles of solute (n) per unit volume (V) of solution.

► See video: VPbur at www.everythingscience.co.za

$$C = \frac{n}{V}$$



For this equation, the units for volume are dm^3 (which is equal to litres). Therefore, the unit of concentration is $\text{mol} \cdot \text{dm}^{-3}$.

DEFINITION: Concentration

Concentration is a measure of the amount of solute that is dissolved in a given volume of liquid. It is measured in $\text{mol} \cdot \text{dm}^{-3}$.

Example 12: Concentration Calculations 1**QUESTION**

If 3,5 g of sodium hydroxide (NaOH) is dissolved in 2,5 dm^3 of water, what is the concentration of the solution in $\text{mol} \cdot \text{dm}^{-3}$?

SOLUTION

Step 1 : **Find the number of moles of sodium hydroxide**

$$n = \frac{m}{M} = \frac{3,5 \text{ g}}{40,01 \text{ g} \cdot \text{mol}^{-1}} = 0,0875 \text{ mol}$$

Step 2 : **Calculate the concentration**

$$C = \frac{n}{V} = \frac{0,0875 \text{ mol}}{2,5 \text{ dm}^3} = 0,035 \text{ mol} \cdot \text{dm}^{-3}$$

The concentration of the solution is $0,035 \text{ mol} \cdot \text{dm}^{-3}$.

Example 13: Concentration Calculations 2**QUESTION**

You have a 1 dm^3 container in which to prepare a solution of potassium permanganate (KMnO_4). What mass of KMnO_4 is needed to make a solution with a concentration of $0,2 \text{ mol} \cdot \text{dm}^{-3}$?

SOLUTION

Step 1 : **Calculate the number of moles**

$$C = \frac{n}{V} \text{ therefore:}$$

$$n = C \times V = 0,2 \text{ mol} \cdot \text{dm}^{-3} \times 1 \text{ dm}^3 = 0,2 \text{ mol}$$

Step 2 : **Find the mass**

$$m = n \times M = 0,2 \text{ mol} \times 158 \text{ g} \cdot \text{mol}^{-1} = 31,6 \text{ g}$$

The mass of KMnO_4 that is needed is 31,6 g.

Example 14: Concentration Calculations 3**QUESTION**

How much sodium chloride (in g) will one need to prepare 500 cm^3 of solution with a concentration of $0,01 \text{ mol} \cdot \text{dm}^{-3}$?

SOLUTION

Step 1 : **Convert the given volume to the correct units**

$$V = 500 \text{ cm}^3 \frac{1 \text{ dm}^3}{1000 \text{ cm}^3} = 0,5 \text{ dm}^3$$

Step 2 : **Find the number of moles**

$$n = C \times V = 0,01 \text{ mol} \cdot \text{dm}^{-3} \times 0,5 \text{ dm}^{-3} = 0,005 \text{ mol}$$

Step 3 : **Find the mass**

$$m = n \times M = 0,005 \text{ mol} \times 58,45 \text{ g} \cdot \text{mol}^{-1} = 0,29 \text{ g}$$

The mass of sodium chloride needed is 0,29 g

Exercise 19 - 6

1. 5,95 g of potassium bromide was dissolved in 400 cm³ of water. Calculate its concentration.
2. 100 g of sodium chloride (NaCl) is dissolved in 450 cm³ of water.
 - a. How many moles of NaCl are present in solution?
 - b. What is the volume of water (in dm³)?
 - c. Calculate the concentration of the solution.
3. What is the molarity of the solution formed by dissolving 80 g of sodium hydroxide (NaOH) in 500 cm³ of water?
4. What mass (g) of hydrogen chloride (HCl) is needed to make up 1000 cm³ of a solution of concentration 1 mol · dm⁻³?
5. How many moles of H₂SO₄ are there in 250 cm³ of a 0,8 mol · dm⁻³ sulphuric acid solution? What mass of acid is in this solution?

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 0080 (2.) 0081 (3.) 0082 (4.) 0083 (5.) 0084

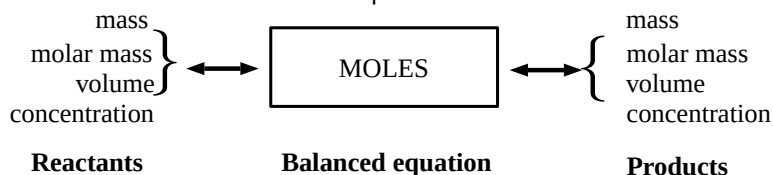
Stoichiometric calculations

 ESAGF

Stoichiometry is the calculation of the quantities of reactants and products in chemical reactions. It is important to know how much product will be formed in a chemical reaction,

or how much of a reactant is needed to make a specific product.

The following diagram shows how the concepts that we have learnt in this chapter relate to each other and to the balanced chemical equation:



▶ See video: VPbxk at www.everythingscience.co.za

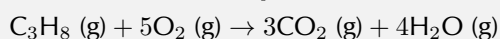
Example 15: Stoichiometric calculation 1

QUESTION

What volume of oxygen at S.T.P. is needed for the complete combustion of 2 dm^3 of propane (C_3H_8)? (Hint: CO_2 and H_2O are the products in this reaction (and in all combustion reactions))

SOLUTION

Step 1 : **Write the balanced equation**



Step 2 : **Find the ratio**

Because all the reactants are gases, we can use the mole ratios to do a comparison. From the balanced equation, the ratio of oxygen to propane in the reactants is 5 : 1.

Step 3 : **Find the answer**

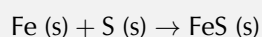
One volume of propane needs five volumes of oxygen, therefore 2 dm^3 of propane will need 10 dm^3 of oxygen for the reaction to proceed to completion.

Example 16: Stoichiometric calculation 2**QUESTION**

What mass of iron (II) sulphide is formed when 5,6 g of iron is completely reacted with sulphur?

SOLUTION

Step 1 : **Write the balanced equation**



Step 2 : **Calculate the number of moles**

We find the number of moles of the given substance:

$$n = \frac{m}{M} = \frac{5,6 \text{ g}}{55,8 \text{ g} \cdot \text{mol}^{-1}} = 0,1 \text{ mol}$$

Step 3 : **Find the mole ratio**

We find the mole ratio between what was given and what you are looking for. From the equation 1 mol of Fe gives 1mol of FeS. Therefore, 0,1 mol of iron in the reactants will give 0,1 mol of iron sulphide in the product.

Step 4 : **Find the mass of iron sulphide**

$$m = n \times M = 0,1 \text{ mol} \times 87,9 \text{ g} \cdot \text{mol}^{-1} = 8,79 \text{ g}$$

The mass of iron (II) sulphide that is produced during this reaction is 8,79 g.

Theoretical yield



When we are given a known mass of a reactant and are asked to work out how much product is formed, we are working out the theoretical yield of the reaction. In the laboratory, chemists almost never get this amount of product. In each step of a reaction a small amount

of product and reactants is “lost” either because a reactant did not completely react or some other unwanted products are formed. This amount of product that you actually got is called the actual yield. You can calculate the percentage yield with the following equation:

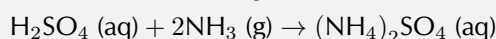
$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

Example 17: Industrial reaction to produce fertiliser**QUESTION**

Sulphuric acid (H_2SO_4) reacts with ammonia (NH_3) to produce the fertiliser ammonium sulphate ($(\text{NH}_4)_2\text{SO}_4$). What is the theoretical yield of ammonium sulphate that can be obtained from 2,0 kg of sulphuric acid? It is found that 2,2 kg of fertiliser is formed. Calculate the % yield.

SOLUTION

Step 1 : **Write the balanced equation**



Step 2 : **Calculate the number of moles of the given substance**

$$n(\text{H}_2\text{SO}_4) = \frac{m}{M} = \frac{2\,000 \text{ g}}{98,12 \text{ g} \cdot \text{mol}^{-1}} = 20,38320424 \text{ mol}$$

Step 3 : **Find the mole ratio**

From the balanced equation, the mole ratio of H_2SO_4 in the reactants to $(\text{NH}_4)_2\text{SO}_4$ in the product is 1 : 1. Therefore, 20,383 mol of H_2SO_4 forms 20,383 mol of $(\text{NH}_4)_2\text{SO}_4$.

Step 4 : **Write the answer**

The maximum mass of ammonium sulphate that can be produced is calculated as follows:

$$m = n \times M = 20,383 \text{ mol} \times 114,04 \text{ g} \cdot \text{mol}^{-1} = 2\,324,477 \text{ g}$$

The maximum amount of ammonium sulphate that can be produced is 2,324 kg.

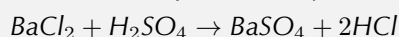
Step 5 : **Calculate the % yield**

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100 = \frac{2,2}{2,324} \times 100 = 94,64\%$$

Example 18: Calculating the mass of reactants and products

QUESTION

Barium chloride and sulphuric acid react according to the following equation to produce barium sulphate and hydrochloric acid.



If you have 2 g of BaCl_2 :

1. What quantity (in g) of H_2SO_4 will you need for the reaction so that all the barium chloride is used up?
2. What mass of HCl is produced during the reaction?

SOLUTION

Step 1 : **Find the number of moles of barium chloride**

$$n = \frac{m}{M} = \frac{2 \text{ g}}{208,2 \text{ g} \cdot \text{mol}^{-1}} = 0,0096 \text{ mol}$$

Step 2 : **Find the number of moles of sulphuric acid**

According to the balanced equation, 1 mole of BaCl_2 will react with 1 mole of H_2SO_4 . Therefore, if 0,0096 mol of BaCl_2 react, then there must be the same number of moles of H_2SO_4 that react because their mole ratio is 1 : 1.

Step 3 : **Find the mass of sulphuric acid**

$$m = n \times M = 0,0096 \text{ mol} \times 98,12 \text{ g} \cdot \text{mol}^{-1} = 0,94 \text{ g}$$

(answer to 1)

Step 4 : **Find the moles of hydrochloric acid**

According to the balanced equation, 2 moles of HCl are produced for every 1 mole of the two reactants. Therefore the number of moles of

HCl produced is $(2 \times 0,0096 \text{ mol})$, which equals $0,0192 \text{ mol}$.

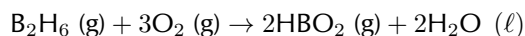
Step 5 : **Find the mass of hydrochloric acid**

$$m = n \times M = 0,0192 \text{ mol} \times 36,46 \text{ g} \cdot \text{mol} = 0,7 \text{ g}$$

(answer to 2)

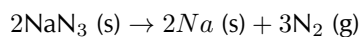
Exercise 19 - 7

1. Diborane, B_2H_6 , was once considered for use as a rocket fuel. The combustion reaction for diborane is:



If we react $2,37 \text{ g}$ of diborane, how many grams of water would we expect to produce?

2. Sodium azide is a commonly used compound in airbags. When triggered, it has the following reaction:



If $23,4 \text{ g}$ of sodium azide is used, how many moles of nitrogen gas would we expect to produce? What volume would this nitrogen gas occupy at STP?

3. Photosynthesis is a chemical reaction that is vital to the existence of life on Earth. During photosynthesis, plants and bacteria convert carbon dioxide gas, liquid water, and light into glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) and oxygen gas.
- Write down the equation for the photosynthesis reaction.
 - Balance the equation.
 - If 3 mol of carbon dioxide are used up in the photosynthesis reaction, what mass of glucose will be produced?

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 0085 (2.) 0086 (3.) 0087

Chapter 19 | Summary

See the summary presentation (📺 Presentation: VPeyf at www.everythingscience.co.za)

- The **mole (n)** (abbreviation mol) is the SI (Standard International) unit for amount of substance.
- The number of particles in a mole is called **Avogadro's number** and its value is $6,022 \times 10^{23}$. These particles could be atoms, molecules or other particle units, depending on the substance.
- The **molar mass (M)** is the mass of one mole of a substance and is measured in grams per mole or $\text{g} \cdot \text{mol}^{-1}$. The numerical value of an element's molar mass is the same as its relative atomic mass. For a covalent compound, the molar mass has the same numerical value as the molecular mass of that compound. For an ionic substance, the molar mass has the same numerical value as the formula mass of the substance.
- The relationship between moles (n), mass in grams (m) and molar mass (M) is defined by the following equation:

$$n = \frac{m}{M}$$

- In a balanced chemical equation, the number in front of the chemical symbols describes the **mole ratio** of the reactants and products.
- The **empirical formula** of a compound is an expression of the relative number of each type of atom in the compound.
- The **molecular formula** of a compound describes the actual number of atoms of each element in a molecule of the compound.
- The formula of a substance can be used to calculate the **percentage by mass** that each element contributes to the compound.
- The **percentage composition** of a substance can be used to deduce its chemical formula.
- We can use the products of a reaction to determine the formula of one of the reactants.
- We can find the number of moles of waters of crystallisation.
- One mole of gas occupies a volume of $22,4 \text{ dm}^3$ at S.T.P..
- The **concentration** of a solution can be calculated using the following equation,

$$C = \frac{n}{V}$$

where C is the concentration (in $\text{mol} \cdot \text{dm}^{-3}$), n is the number of moles of solute dissolved in the solution and V is the volume of the solution (in dm^{-3}). The concentration is a measure of the amount of solute that is dissolved in a given volume of liquid.

- The concentration of a solution is measured in $\text{mol} \cdot \text{dm}^{-3}$.
- **Stoichiometry** is the calculation of the quantities of reactants and products in chemical reactions. It is also the numerical relationship between reactants and products.
- The theoretical yield of a reaction is the maximum amount of product that we expect

to get out of a reaction.

Chapter 19

End of chapter exercises

1. Write only the word/term for each of the following descriptions:
 - a. the mass of one mole of a substance
 - b. the number of particles in one mole of a substance
2. 5 g of magnesium chloride is formed as the product of a chemical reaction. Select the **true** statement from the answers below:
 - a. 0,08 moles of magnesium chloride are formed in the reaction
 - b. the number of atoms of Cl in the product is $0,6022 \times 10^{23}$
 - c. the number of atoms of Mg is 0,05
 - d. the atomic ratio of Mg atoms to Cl atoms in the product is 1 : 1
3. 2 moles of oxygen gas react with hydrogen. What is the mass of oxygen in the reactants?
 - a. 32 g
 - b. 0,125 g
 - c. 64 g
 - d. 0,063 g
4. In the compound potassium sulphate (K_2SO_4), oxygen makes up $x\%$ of the mass of the compound. $x = ?$
 - a. 36,8
 - b. 9,2
 - c. 4
 - d. 18,3
5. The concentration of a 150 cm³ solution, containing 5 g of NaCl is:
 - a. $0,09 \text{ mol} \cdot \text{dm}^{-3}$
 - b. $5,7 \times 10^{-4} \text{ mol} \cdot \text{dm}^{-3}$
 - c. $0,57 \text{ mol} \cdot \text{dm}^{-3}$
 - d. $0,03 \text{ mol} \cdot \text{dm}^{-3}$
6. Calculate the number of moles in:
 - a. 5 g of methane (CH_4)
 - b. 3,4 g of hydrochloric acid
 - c. 6,2 g of potassium permanganate (KMnO_4)
 - d. 4 g of neon
 - e. 9,6 kg of titanium tetrachloride (TiCl_4)
7. Calculate the mass of:
 - a. 0,2 mol of potassium hydroxide (KOH)
 - b. 0,47 mol of nitrogen dioxide

- c. 5,2 mol of helium
 - d. 0,05 mol of copper (II) chloride (CuCl_2)
 - e. $31,31 \times 10^{23}$ molecules of carbon monoxide (CO)
8. Calculate the percentage that each element contributes to the overall mass of:
- a. Chloro-benzene ($\text{C}_6\text{H}_5\text{Cl}$)
 - b. Lithium hydroxide (LiOH)
9. CFC's (chlorofluorocarbons) are one of the gases that contribute to the depletion of the ozone layer. A chemist analysed a CFC and found that it contained 58,64% chlorine, 31,43% fluorine and 9,93% carbon. What is the empirical formula?
10. 14 g of nitrogen combines with oxygen to form 46 g of a nitrogen oxide. Use this information to work out the formula of the oxide.
11. Iodine can exist as one of three oxides (I_2O_4 ; I_2O_5 ; I_4O_9). A chemist has produced one of these oxides and wishes to know which one they have. If he started with 508 g of iodine and formed 652 g of the oxide, which oxide has he produced?
12. A fluorinated hydrocarbon (a hydrocarbon is a chemical compound containing hydrogen and carbon) was analysed and found to contain 8,57% H, 51,05% C and 40,38% F.
- a. What is its empirical formula?
 - b. What is the molecular formula if the molar mass is $94,1 \text{ g} \cdot \text{mol}^{-1}$?
13. Copper sulphate crystals often include water. A chemist is trying to determine the number of moles of water in the copper sulphate crystals. She weighs out 3 g of copper sulphate and heats this. After heating, she finds that the mass is 1,9 g. What is the number of moles of water in the crystals? (Copper sulphate is represented by $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$).
14. 300 cm^3 of a $0,1 \text{ mol} \cdot \text{dm}^{-3}$ solution of sulphuric acid is added to 200 cm^3 of a $0,5 \text{ mol} \cdot \text{dm}^{-3}$ solution of sodium hydroxide.
- a. Write down a balanced equation for the reaction which takes place when these two solutions are mixed.
 - b. Calculate the number of moles of sulphuric acid which were added to the sodium hydroxide solution.
 - c. Is the number of moles of sulphuric acid enough to fully neutralise the sodium hydroxide solution? Support your answer by showing all relevant calculations.
15. A learner is asked to make 200 cm^3 of sodium hydroxide (NaOH) solution of concentration $0,5 \text{ mol} \cdot \text{dm}^{-3}$.
- a. Determine the mass of sodium hydroxide pellets he needs to use to do this.
 - b. Using an accurate balance the learner accurately measures the correct mass of the NaOH pellets. To the pellets he now adds exactly 200 cm^3 of pure water. Will his solution have the correct concentration? Explain your answer.

- c. The learner then takes 300 cm^3 of a $0,1 \text{ mol} \cdot \text{dm}^{-3}$ solution of sulphuric acid (H_2SO_4) and adds it to 200 cm^3 of a $0,5 \text{ mol} \cdot \text{dm}^{-3}$ solution of NaOH at 25°C .
- d. Write down a balanced equation for the reaction which takes place when these two solutions are mixed.
- e. Calculate the number of moles of H_2SO_4 which were added to the NaOH solution.
16. $96,2 \text{ g}$ sulphur reacts with an unknown quantity of zinc according to the following equation: $\text{Zn} + \text{S} \rightarrow \text{ZnS}$
- a. What mass of zinc will you need for the reaction, if all the sulphur is to be used up?
- b. Calculate the theoretical yield for this reaction.
- c. It is found that 275 g of zinc sulphide was produced. Calculate the % yield.
17. Calcium chloride reacts with carbonic acid to produce calcium carbonate and hydrochloric acid according to the following equation:
 $\text{CaCl}_2 + \text{H}_2\text{CO}_3 \rightarrow \text{CaCO}_3 + 2\text{HCl}$
If you want to produce 10 g of calcium carbonate through this chemical reaction, what quantity (in g) of calcium chloride will you need at the start of the reaction?

 More practice  video solutions  or help at www.everythingscience.co.za

- (1.) 0088 (2.) 0089 (3.) 008a (4.) 008b (5.) 008c (6.) 008d
(7.) 008e (8.) 008f (9.) 008g (10.) 008h (11.) 008i (12.) 008j
(13.) 008k (14.) 008m (15.) 008n (16.) 008p (17.) 008q